1) Lesson 9/10/26

1.1) Inequalities

General Procedure:

1. Determine the domain D.
2. Bring all terms to one side: $f(x) \gtrless 0$.
3. Factor into products or fractions: $\frac{A(x)}{B(x)} \gtrless 0$.
    1. Find the roots of each factor.
    2. Construct a sign chart.
4. Intersect the resulting intervals with the domain D.

There can happen 3 main cases: fractions, square roots or absolute values

Ex c) Fraction

Solve the following inequality:

$$\frac{x-1}{x-2} > \frac{2x-3}{x-3}$$

Domain:

$$D = x \in \mathbb{R} \setminus \{2, 3\}$$

Simplify:

$$\frac{x-1}{x-2} - \frac{2x-3}{x-3} > 0 \implies \frac{(x-1)(x-3) - (2x-3)(x-2)}{(x-2)(x-3)} > 0$$

This is of form $A(x)/B(x)$ so the study of the numerator and denominator needs to be done independently

Numerator:
Find zeroes:

$$\begin{aligned}(x-1)(x-3) - (2x-3)(x-2) &= (x^2 - 4x + 3) - (2x^2 - 7x + 6) \\ &= -x^2 + 3x - 3\end{aligned}$$

Recall the quadratic formula:
Let $a, b, c \in \mathbb{R}$, then $ax^2 + bx + c = 0$ has the following solution:

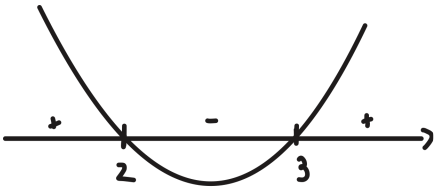
$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Where the discriminant $\Delta = b^2 - 4ac$ determines the types of solution:

- $\Delta > 0$: two real (distinct) solutions
- $\Delta = 0$: two real (identical) solutions
- $\Delta < 0$: no real solution

In this case since $\Delta = 9 - 12 = -3 < 0$ we don't have any solutions, hence since $a < 0$ the whole numerator is negative.

Denominator:
Here the function already shows the zeroes, so we can plot the graph to see the regions:



Curve

So we have positive denominator for $x < 2 \cup x > 3$.

By noticing the interval where they have the same sign:

x	$(-\infty, 2)$	$(2, 3)$	$(3, +\infty)$
Numerator	$-$	$-$	$-$
Denominator	$+$	$-$	$+$
Fraction	$-$	$+$	$-$

Finally we have:

$$\boxed{2 < x < 3}$$

Ex d) Absolute value

Solve the following inequality:

$$\frac{|x|}{x-1} > \frac{x+1}{2x-1}$$

Domain:

$$D = x \in \mathbb{R} \setminus \left\{ \frac{1}{2}, 1 \right\}$$

Now recall the abs value definition:

$$|f(x)| = \begin{cases} f(x) & f(x) \geq 0 \\ -f(x) & f(x) < 0 \end{cases}$$

In this case $f(x) = x$ so we have:

$$\begin{cases} \dfrac{x}{x-1} > \dfrac{x+1}{2x-1} & x \geq 0 \\ \dfrac{-x}{x-1} > \dfrac{x+1}{2x-1} & x < 0 \end{cases}$$

Remeber:

- The solution to each part of the system must be put in intersection with the condition
- The solutions of the parts of a system are in union with each other

Start with case 1:
Rewrite so that it is a single fraction:

$$\frac{x}{x-1} - \frac{x+1}{2x-1} > 0 \implies \frac{x(2x-1)-(x+1)(x-1)}{(x-1)(2x-1)} > 0$$

- Numerator:

$$x(2x-1)-(x^2-1) = 2x^2 - x - x^2 + 1 = x^2 - x + 1$$

Find the zeroes of the numerator:

$$x^2 - x + 1 = 0 \to \Delta = -3$$

Since $\Delta < 0$ and $a = 1 > 0$, the numerator is strictly positive.

- Denominator:

$$(x-1)(2x-1) > 0 \iff x < \frac{1}{2} \cup x > 1$$

Sign analysis for Case 1:

x	$\left(-\infty, \frac{1}{2}\right)$	$\left(\frac{1}{2}, 1\right)$	$(1, +\infty)$
Numerator	$+$	$+$	$+$
Denominator	$+$	$-$	$+$
Fraction	$+$	$-$	$+$

So the fraction is > 0 on $\left(-\infty, \frac{1}{2}\right) \cup (1, +\infty)$.

Intersecting with the case condition $x \geq 0$:

$$S_1 = \left[0, \frac{1}{2}\right) \cup (1, +\infty)$$

Case 2:
Same procedure as case 1:

$$\frac{-x}{x-1} - \frac{x+1}{2x-1} > 0 \implies \frac{-x(2x-1)-(x+1)(x-1)}{(x-1)(2x-1)} > 0$$

- Numerator:

$$-2x^2 + x - (x^2 - 1) = -3x^2 + x + 1$$

Find the zeroes of the numerator:

$$x_{1,2} = \frac{-1 \pm \sqrt{13}}{-6} = \frac{1 \mp \sqrt{13}}{6}$$

Since $a = -3 < 0$, the parabola opens downward:

$$N > 0 \iff x_1 = \frac{1-\sqrt{13}}{6} < x < \frac{1+\sqrt{13}}{6} = x_2$$

Denominator:
The denominator roots are $\frac{1}{2} = 0.5$ and 1.

Sign analysis for the factors:

x	$(-\infty, x_1)$	$(x_1, 1/2)$	$(1/2, x_2)$	$(x_2, 1)$	$(1, +\infty)$
Numerator	$-$	$+$	$+$	$-$	$-$
Denominator	$+$	$+$	$-$	$-$	$+$
Fraction	$-$	$+$	$-$	$+$	$-$

The fraction is > 0 on $\left(x_1, \frac{1}{2}\right) \cup (x_2, 1)$.

Intersecting with the case condition $x < 0$:

$$S_2 = \left(\frac{1-\sqrt{13}}{6}, 0\right)$$

Taking the union of both cases $S = S_1 \cup S_2$:

$$S = \left(\frac{1-\sqrt{13}}{6}, 0\right) \cup \left[0, \frac{1}{2}\right) \cup (1, +\infty)$$

Notice that 0 is included in the solution set, so the first two intervals join together:

$$\boxed{S = \left(\frac{1-\sqrt{13}}{6}, \frac{1}{2}\right) \cup (1, +\infty)}$$

Ex f) Square Root

Solve the following inequality:

$$\sqrt{x^2 - 6x} > x + 2$$

First notice that the domain is

$$x^2 - 6x \geq 0 \rightarrow D = (-\infty, 0] \cup [6, +\infty)$$

First notice that when $x + 2 < 0$ the solution inequality holds (in the domain). So one part of the solution is:

$$S_1 = D \cap \{x < -2\} = \{x < -2\}$$

Now for $x + 2 \geq 0$ of both sides can be squared:

$$x^2 - 6x > x^2 + 4x + 4 \rightarrow 0 > 10x + 4 \rightarrow x < -\frac{2}{5}$$

This clearly for the values bigger than -2 in the domain, that is

$$S_2 = D \cap \left\{\{x \geq -2\} \cap \left\{x < -\frac{2}{5}\right\}\right\} = D \cap \left\{-2 \leq x < -\frac{2}{5}\right\} = \left\{-2 \leq x < -\frac{2}{5}\right\}$$

So the result is:

$$S = S_1 \cup S_2 = \{x < -2\} \cup \left\{-2 \leq x < -\frac{2}{5}\right\} = x < -\frac{2}{5}$$

Ex i) Mixed Together

Solve the following inequality:

$$\sqrt{|x^2-4|}-x\geq 0$$

Domain: $|x^2-4|\geq 0 \rightarrow D=\mathbb{R}$

Now recall the abs value definition:

$$|f(x)|=\begin{cases} f(x) & f(x)\geq 0 \\ -f(x) & f(x)<0 \end{cases}$$

In this case $f(x)=x^2-4=(x+2)(x-2)$ so $f(x)\geq 0 \rightarrow x\leq -2 \cup x\geq 2$

Now we can rewrite the inequality as a system of two inequalities:

$$\sqrt{|x^2-4|}-x\geq 0 \rightarrow \begin{cases} \sqrt{x^2-4}-x\geq 0 & x\leq -2 \cup x\geq 2 \\ \sqrt{4-x^2}-x\geq 0 & -2<x<2 \end{cases}$$

Each system can be solved independently from the other.
Call the conditions of the cases D_1 and D_2 respectively.

Solve first case:
Recall that when RHS (x) is negative there is always a solution.

$$\sqrt{x^2-4}\geq x$$

Therefore one part of the solution includes $x<0$

Now study in the interval $x\geq 0$: since we are sure that both sides are ≥ 0 we can square both sides:

$$x^2-4\geq x^2 \rightarrow -4\geq 0$$

Not possible, so the only solution to this part is when x is negative:

$$S_1=x\leq 0 \cap D_1=x\leq -2$$

Solve the second equation:
As before a solution always exists for $x<0$. Now consider only the positive values ($x\geq 0$) in order to be able to square both sides:

$$\sqrt{4-x^2}\geq x \rightarrow 4-x^2\geq x^2 \rightarrow 4-2x^2\geq 0 \rightarrow -\sqrt{2}\leq x\leq \sqrt{2}$$

So we have $S_2=D_2\cap\left\{x<0\cup -\sqrt{2}\leq x\leq \sqrt{2}\right\}=D_2\cap x\leq \sqrt{2}=-2<x\leq \sqrt{2}$

The result is therefore the union of both cases which yields:

$$\boxed{S=x\leq -2\cup -2<x\leq \sqrt{2}\rightarrow S=x\leq \sqrt{2}}$$

1.2) Subsets of $\mathbb{R}$

Ex a)

Describe the following set:

$$A=S_1\cap S_2=\left\{x\in\mathbb{R}:x^2+4x+13<0\right\}\cap\left\{x\in\mathbb{R}:3x^2+5>0\right\}$$

Subset 1:

$$x^2 + 4x + 13 < 0 \to \Delta = -36 \implies S_1 = \emptyset$$

any intersection with an empty set is an empty set:

$$\boxed{A = \emptyset \cap S_2 = \emptyset}$$

In general recall:

$$\emptyset \cap S = \emptyset \qquad \emptyset \cup S = S$$

Ex c)

Describe the following set:

$$C = \left\{ x \in \mathbb{R} : \frac{x^2 - 5x + 4}{x^2 - 9} < 0 \right\} \cup \left\{ x \in \mathbb{R} : x + \sqrt{7x + 1} = 17 \right\}$$

Subset 1:
First notice that the domain is $x \neq \pm 3$

Numerator:

$$x^2 - 5x + 4 = 0 \to x_{1,2} = 1,\ 4 \to N > 0 \iff x < 1 \cup x > 4$$

Denominator:

$$x^2 - 9 = (x + 3)(x - 3) = 0 \to D > 0 \iff x < -3 \cup x > 3$$

x	$(-\infty, -3)$	$(-3, 1)$	$(1, 3)$	$(3, 4)$	$(4, +\infty)$
Numerator	$+$	$+$	$-$	$-$	$+$
Denominator	$+$	$-$	$-$	$+$	$+$
Fraction	$+$	$-$	$+$	$-$	$+$

so $S_1 = (-3, 1) \cup (3, 4)$

Subset 2:
This is just an equality:

$$\sqrt{7x + 1} = 17 - x$$

The domain is $7x + 1 \geq 0 \to D = \left\{ x \geq -\frac{1}{7} \right\}$
Moreover as before, both sides must have the same sign when squaring and since the sqrt is always ≥ 0 we must have that also $17 - x \geq 0 \to x \leq 17$ so we can further reduce the domain to

$$D = \left[-\frac{1}{7}, 17 \right]$$

Square both sides:

$$7x + 1 = (17 - x)^2 \to 7x + 1 = x^2 - 34x + 17^2 \to x^2 - 41x + 288 = 0$$

$$x_{1,2} = 9,\ 32$$

But only 9 is acceptable as $32 \notin D$

Finally:

$$\boxed{C = (-3, 1) \cup (3, 4) \cup \{9\}}$$

1.3) Max/Min, Sup/Inf

Recall the theory

Definition (Upper/Lower Bound)**.**

$M \in \mathbb{Q}$ is called **upper bound** for A if $M \geq a, \forall a \in A$.

Respectively it is called **lower bound** if $M \leq a, \forall a \in A$

Notice M does not need to be in A
Example: consider set $A = [0, 1)$ any $m \leq 0$ is a lower bound and any $M \geq 1$ is upper bound

Definition (Maximum/Minimum of A)**.**

If $\overline{M}$ is an upper bound for A and $\overline{M} \in A$, then $\overline{M}$ is the *minimum* possible *upper* bound and is the **maximum** (of A).

If $\overline{m}$ is an lower bound for A and $\overline{m} \in A$, then $\overline{m}$ is the *maximum* possible *lower* bound and the **minimum** (of A).

In this case $\overline{M} \in A$ by definition.
From the previous example: $\overline{m} = \max\{m \leq 0\} = 0 \in A$ is the correct minimum of A, however $\overline{M} = \min\{M \geq 1\} = 1 \notin A$ means that we don't have a maximum

Definition (Supremum/Infimum)**.**

The **supremum of** A $\sup A$ is the minimum of the upper bounds of A.

The **infimum of** A $\inf A$ is the maximum of the lower bounds of A.

Finish the example: we know $\overline{m} = 0 \in A$ so this is both minimum and infimum of A. We know also $\overline{M} = 1 \notin A$, therefore $\sup \mathrm{A} = 1$ but it is not the maximum.

Recall that if a set is unbounded, then

$$\sup A = +\infty \quad \text{if } A \text{ is unbounded from above}$$

$$\inf A = -\infty \quad \text{if } A \text{ is unbounded from below}$$

Ex a)

Tell wether this subsets is bounded and the sup/inf and if it allow for maximum/minimum.

$$A = \left\{ x \in \mathbb{R} : x = n \text{ or } x = \frac{1}{n^2}, n \in N \setminus \{0\} \right\}$$

It is immediately clear that A is the union of two sets, mainly

$$A = S_1 \cup S_2 \text{ where } \begin{array}{l} S_1 = \{n \in \mathbb{N} \setminus \{0\}\} = \{1, 2, 3, 4, \ldots\} \\ S_2 = \left\{ \frac{1}{n^2} \mid n \in \mathbb{N} \setminus \{0\} \right\} = \left\{ 1, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \ldots \right\} \end{array}$$

Find the bounds on both sets individually (not required but the result is clearer).

Study the first subset S_1: notice that it is just the natural numbers with zero excluded, so its properties are already known.

- No upper bound, so also no max
  - Supremum: since unbounded; sup $S_1 = +\infty$
- Lower bound: $m_1 \leq 1$
  - Minimum: $\overline{m}_1 = \max m_1 \leq 1 = 1 \in S$

Study the second subset S_2: Intuitively we know that the function describing the set is monotonically decreasing: we can express it by saying:

Call x_n the value of x for any chosen n, then $x_{n+1} < x_n$.
This is easily shown as the following inequality is true for $n \in \mathbb{N} \setminus \{0\}$ (check at home):

$$\frac{1}{(n+1)^2} < \frac{1}{n^2}$$

Therefore for every number $x_n > 0$ we can find a smaller number such that $0 < x_{n+1} < x_n$

Is 0 the maximum lower bound?
By contradiction suppose there exists a lower bound $\delta > 0$ then if 0 is the maximum lower bound we have

$$\exists n \in \mathbb{N} \setminus \{0\} \quad \text{such that} \quad \frac{1}{n^2} < \delta$$

Now compute

$$\frac{1}{n^2} < \delta \to n^2 > \frac{1}{\delta} \to n > \frac{1}{\sqrt{\delta}}$$

Because $\delta > 0$, the number $\frac{1}{\sqrt{\delta}}$ is just a fixed positive real number and an appropriate n exists.

With this digression aside, we also know that $0 \notin S_2$.
As before analyze:

- Lower bound: $m_2 \leq 0$
  - No minimum: $\overline{m}_2 = \max m_2 \leq 0 = 0 \notin S_2$
  - Infimum is $\inf S_2 = 0$
- Upper bound: $M_2 \geq 1$
  - $\overline{M}_2 = \min M_2 \geq 1 = 1 \in S_2$

Putting all together:
Now consider the union, so:

- No upper bound since S_1 is unbounded
  - therefore sup $A = +\infty$

- Lower bound: The infimum is the **maximum** of all lower bounds: $\inf A = \min(\inf S_1, \inf S_2) = \min(1, 0) = 0$. Since $0 \notin A$, $\min A$ does not exist.

2) Lesson 16/10/26

2.1) Domains

Recall:

$$f(x) = \begin{matrix} \dfrac{a(x)}{g(x)} & \text{dom } f = \{g(x) \neq 0\} \\ \sqrt{g(x)} & \text{dom } f = \{g(x) \geq 0\} \\ \log(g(x)) & \text{dom } f = \{g(x) > 0\} \end{matrix}$$

a) Fraction and Sqrt

Determine the domains of the following function:

$$f(x) = \frac{\sqrt{x^2 - 3x - 4}}{x + 5}$$

Here the domain is given by two solutions that must hold simultaneously.

Fraction:
Here $D_1 = \{x + 5 \neq 0\} = \{x \neq -5\}$

Sqrt:
Here $D_2 = \{x^2 - 3x - 4 \geq 0\}$. The zeroes of this function are $x_{1,2} = -1, 4$ so the inequality holds for $D_2 = \{x \leq -1 \cup x \geq 4\}$

Finally

$$\text{dom } f = D_1 \cap D_2 = \{x \leq -1 \cup x \geq 4\} \setminus \{-5\} = (-\infty, -5) \cup (-5, -1] \cup [4, +\infty)$$

where the last part is the correct formal way to write the result.

b) Logarithm

Determine the domains of the following function:

$$f(x) = \log(x^2 - x)$$

The domain is given by $x^2 - x > 0$, so solve the inequality:

$$x^2 - x = x(x - 1) > 0$$

The first term is positive for $x > 0$, the second for $x > 1$, so

$$\text{dom } f = \{x < 0 \cup x > 1\}$$

c) System

Determine the domains of the following function:

$$f(x) = \begin{cases} \dfrac{1}{2x + 1} & x \geq 0 \\ e^{\sqrt{x+1}} & x < 0 \end{cases}$$

Here the domain is given by the union of the two cases.

Case 1:
Here the domain is given by $D_1^* = \{2x + 1 \neq 0\} = \left\{x \neq -\frac{1}{2}\right\}$. However since this is a system, this must hold

only inside the specified case, so $D_1 = D_1^* \cap \{x \geq 0\} = \{x \geq 0\}$ since $-\frac{1}{2} \notin \{x \geq 0\}$, so this part of the map is always valid.

Case 2:
Here the domain is given by $x + 1 \geq 0 \rightarrow x \geq -1$ which combined with the system is $x \in [-1, 0)$

The final domain becomes the union of the two cases, so

$$\text{dom } f = \{x \geq 0\} \cup \{-1 \leq x < 0\} = \{x \geq -1\}$$

2.2) Symmetries and Periodicity

If a function is symmetric, it can have either even or odd symmetry:

- Even Symmetry: mirrored along y axis: $f(x) = f(-x)$
- Odd Symmetry: mirrored along x and y axis: $f(x) = -f(-x)$

a) Sqrt and Abs

Sketch the graph of the following function indicating the possible symmetries and/or periodicity:

$$f(x) = \sqrt{1 - |x|}$$

Domain:

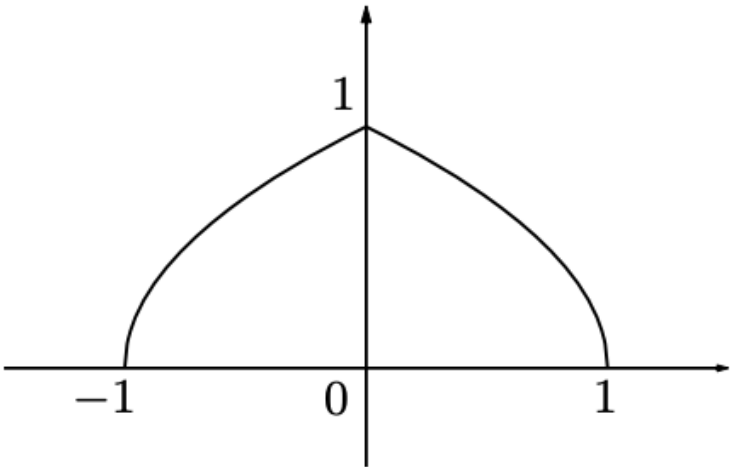
$$D = \{1 - |x| \geq 0\} = \{|x| \leq 1\} = \{x \in [-1, 1]\}$$

So clearly the function cannot be periodic. By definition, a periodic function requires $x \in \text{dom } f \implies x + T \in \text{dom } f$ for all x, which cannot hold on a bounded interval

Symmetry:
Since f is the result of a sqrt, it can only be non negative (roots at $x = \pm 1$), therefore an odd symmetry is not possible (only possible if f is constant 0). Try the even symmetry:

$$f(-x) = \sqrt{1 - |-x|} = \sqrt{1 - |-1| \cdot |x|} = \sqrt{1 - 1 \cdot |x|} = f(x)$$

The final sketch can be done by sampling points only between $[0, 1]$ since then it can be mirrored.



Sketch

b) Cosine

Sketch the graph of the following function indicating the possible symmetries and/or periodicity:

$$f(x) = 1 + \cos(2x)$$

There is no restriction in the domain, so start with the symmetries. Again since $\cos(z) \in [-1, 1]$ the function returns only non-negative values, so an odd symmetry is excluded, moreover since $\cos(z)$ is an even function, we can guess that the function is even, let's prove it:

$$f(-x) = 1 + \cos(-2x) = 1 + \cos(2x) = f(x)$$

Periodicity:
T_1: 1 is a constant function. A constant function $c(x) = c$ satisfies $c(x + T) = c(x)$ for **every** $T > 0$. Therefore, every positive real number is a period. However it does not have a fundamental period (0 cannot be period)
T_2: $\cos(\theta)$ has period 2π, that is $\cos(\theta + 2\pi) = \cos(\theta)$, but what about $g(x) = \cos(2x)$? Let's try the generic case $\cos(ax)$, $a > 0$

By the definition of period, we must find a $T > 0$ such that

$$\min_T \{T > 0 : g(x + T) = g(x)\ \forall x \in \mathbb{R}\} = \min_T \{\cos(a(x + T)) = \cos(ax)\}$$

Recall that $\cos(\theta + 2\pi) = \cos(\theta)$ and by comparing it with the argument of the minimum we get

$$aT = 2\pi \to T = \frac{2\pi}{a}$$

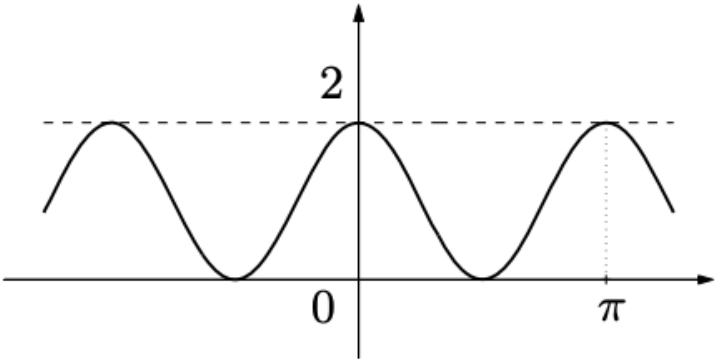
And in our case $T_2 = \pi$

Adding a constant is merely a vertical translation and leaves the periodic properties unchanged:

$$f(x + T) = f(x) \iff 1 + \cos(2(x + T)) = 1 + \cos(2x) \iff \cos(2(x + T)) = \cos(2x)$$

Therefore, the fundamental period of $f(x)$ is simply the fundamental period of $\cos(2x)$:

$$T = \frac{2\pi}{2} = \pi$$



Sketch

Two sinusoids

Determine the period of the following function:

$$f(x) = \sin(2x) + \cos(\pi x)$$

From the previous exercise we know that a sinusoidal has period

$$T = \frac{2\pi}{a}$$

So we know the two periods:

$$T_1 = \frac{2\pi}{2} = \pi \qquad T_2 = \frac{2\pi}{\pi} = 2$$

Now for the final period, recall that **the sum of two periodic functions is periodic if and only if the ratio of their periods is a rational number.**

Check the ratio:

$$\frac{T_1}{T_2} = \frac{\pi}{2} \notin \mathbb{Q}$$

so the function is aperiodic.

Now consider

$$f(x) = \sin(2x) + \cos(3x)$$

therefore

$$\begin{aligned} T_1 &= \pi \\ T_2 &= \frac{2\pi}{3} \end{aligned} \qquad \frac{T_1}{T_2} = \frac{3}{2} \in \mathbb{Q}$$

We know the function is periodic! But how is the period found?
To find the fundamental period T, any common period must satisfy $f(x+T) = f(x)$ for all x. For the sum of two independent sinusoids, this requires T to be a simultaneous integer multiple of both individual periods T_1 and T_2:

$$T = k_1 T_1 = k_2 T_2 \quad \text{for some } k_1, k_2 \in \mathbb{Z}^+$$

That is:

$$k_1 \pi = k_2 \left(\frac{2\pi}{3}\right) \implies k_1 = \frac{2}{3} k_2 \implies \frac{k_1}{k_2} = \frac{2}{3} \qquad k_1 = 2, k_2 = 3$$

and thus the period is

$$T = 2\pi = 3 \cdot \frac{2\pi}{3}$$

1) Arcsin

Determine domain, sign and possible symmetries and periodicity of the following function:

$$f(x) = \arcsin\left(\left|x^3 - \frac{1}{2}\right|\right)$$

Properties of $\arcsin(t)$
The sin function is not bijective (surjective but not injective) the map is $\mathbb{R} \to [-1, 1]$. So the arcsin is defined on the restriction in which this function is strictly increasing and bijective, that is

$$\sin\Big|_{\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]} : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \longrightarrow [-1, 1] \implies \arcsin : [-1, 1] \longrightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

The domain is therefore

$$\text{dom } \arcsin = [-1, 1]$$

The sign coincides with the sign of the argument.

Domain:
Recall that $\arcsin(t)$ is defined for $t \in [-1, 1]$, so we must impose this as the domain:

$$-1 \leq \left|x^3 - \frac{1}{2}\right| \leq 1$$

Since the absolute value is always non-negative we have

$$\left|x^3 - \frac{1}{2}\right| \leq 1 \to -1 \leq x^3 - \frac{1}{2} \leq 1 \to -\frac{1}{\sqrt[3]{2}} \leq x \leq \sqrt[3]{\frac{3}{2}}$$

So we have

$$\operatorname{dom} f = \left[-\frac{1}{\sqrt[3]{2}}, \sqrt[3]{\frac{3}{2}} \right]$$

Here we can notice that since the domain is not symmetric wrt the origin, the function does not have any symmetries, that is: A function can have symmetry **only if** its domain is symmetric with respect to the origin

$$x \in \operatorname{dom} f \implies -x \in \operatorname{dom} f$$

Moreover, since it has a finite domain it cannot be periodic.

Sign:
In our case since we have only positive values, the function will always be positive

$$\arcsin(t) \in \left[0, \frac{\pi}{2}\right] \implies \arcsin(t) \geq 0$$

The root is at $\left|x^3 - \frac{1}{2}\right| = 0 \to x = \sqrt[3]{\frac{1}{2}}$

3) Exponential

Determine domain, sign and possible symmetries and periodicity of the following function:

$$f(x) = \log(e^{2x} - 4e^x + 4)$$

First we do a substitution, $y \triangleq e^x$ and therefore

$$f(y) = \log(y^2 - 4y + 4)$$

Domain:
From the log we impose $y^2 - 4y + 4 > 0$
The roots are $y_{1,2} = 2$ which for x becomes:

$$\begin{aligned} 2 &= e^x \\ \log 2 &= \log e^x \\ x &= \log 2 \end{aligned}$$

so the domain is just dom $f = \mathbb{R} \setminus \{\log 2\} = (-\infty, \log 2) \cup (\log 2, +\infty)$
Since domain is not symmetric wrt to origin, we don't have any symmetry.

Sign:
Recall $\log(z) \geq 0 \iff z \geq 1$, so

$$y^2 - 4y + 4 \geq 1 \to y^2 - 4y + 3 \geq 0$$

The roots are $y_{1,2} = 1, 3 \to x_{1,2} = 0, \log 3$
The inequality holds for $x \leq 0 \cup x \geq \log 3$

So $f(x) \geq 0 \iff x \leq 0$ or $x \geq \log 3$
And the roots are at $x = 0,\ x = \log 3$

7) Hyperbolic Cosine

Determine domain, sign and possible symmetries and periodicity of the following function:

$$f(x) = \frac{1}{\cosh(\sin(x))}$$

What is the hyperbolic cosine?

$$\cosh(t) = \frac{e^t + e^{-t}}{2}$$

So this is always defined and is always positive.
This is also an even function:

$$\cosh(-t) = \frac{e^{-t} + e^t}{2} = \cosh(t)$$

Domain:
First define $a = \sin(x) \in [-1, 1]$
The denominator cannot be 0 so

$$\cosh(a) = \frac{e^a + e^{-a}}{2} \neq 0$$

Which is valid $\forall a$ as $e^k > 0 \quad \forall k \in \mathbb{R}$.

So

$$\text{dom } f = \mathbb{R}$$

Sign:
Hyperbolic cosine is always positive so also this function is always positive with no zero

Symmetry:
Cosh is even $\cosh(x) = \cosh(-x)$, sin is odd $\sin(-x) = -\sin(x)$.
Try even:

$$f(-x) = \frac{1}{\cosh(\sin(-x))} = \frac{1}{\cosh(-\sin(x))} = \frac{1}{\cosh(\sin(x))} = f(x)$$

So it is even!

Is it periodic?
As before impose

$$f(x + T) = f(x) \quad \text{for all } x \in \mathbb{R}$$

Substituting

$$\frac{1}{\cosh(\sin(x + T))} = \frac{1}{\cosh(\sin(x))} \to \cosh(\sin(x + T)) = \cosh(\sin(x))$$

Easy way: Try period of $\sin(x)$ (2π):

$$\begin{aligned} \cosh(\sin(x + 2\pi)) &= \cosh(\sin(x)) \\ \cosh(\sin(x)) &= \cosh(\sin(x)) \end{aligned}$$

Harder way:
Since cosh has even symmetry:

$$\cosh(t) = \cosh(-t) \implies \cosh(t) = \cosh(|t|) \quad \forall t \in \mathbb{R}$$

And therefore (prove at home):

$$\cosh(|u|) = \cosh(|v|) \iff |u| = |v|$$

And thus impose:

$$|\sin(x + T)| = |\sin(x)| \quad \forall x \in \mathbb{R}$$

Squaring both sides:

$$\sin^2(x+T) = \sin^2(x)$$

Using the identity $\sin^2(\theta) = \frac{1-\cos(2\theta)}{2}$:

$$\frac{1-\cos(2x+2T)}{2} = \frac{1-\cos(2x)}{2} \iff \cos(2x+2T) = \cos(2x) \quad \forall x \in \mathbb{R}$$

Therefore the period is the same as $\cos(2x)$ which is $T = \pi$ and thus 2π is not the fundamental period (it is a period nonetheless).

10)

Determine domain, sign and possible symmetries and periodicity of the following function:

$$f(x) = \frac{\log(\sin(x))}{\sin(x) - 1}$$

Domain:
Denominator forces $\sin(x) \neq 1$ which holds for $x \neq \frac{\pi}{2} + 2k\pi \quad k \in \mathbb{Z}$
The log forces $\sin(x) > 0$ which is true for $x \in (2k\pi, \pi + 2k\pi) \quad k \in \mathbb{Z}$
so the generic k-th interval is $(2k\pi, \frac{\pi}{2} + 2k\pi) \cup (\frac{\pi}{2} + 2k\pi, \pi + 2k\pi)$
The domain is the union of all these intervals:

$$\text{dom } f = \bigcup_{k \in \mathbb{Z}} (2k\pi, \frac{\pi}{2} + 2k\pi) \cup (\frac{\pi}{2} + 2k\pi, \pi + 2k\pi)$$

which is not symmetric wrt the origin and thus there is no symmetry

Sign:
The denominator is always negative since $\max \sin(x) = 1$ and thus $\max \sin(x) - 1 = 0$ (which is excluded by domain)
The numerator is also always negative:

$$\log(\sin(x)) \geq 0 \iff \sin(x) \geq 1 \iff \sin(x) = 1$$

which is excluded from the domain

Therefore this function is always positive

Periodicity:
Notice that the domain itself needs to be
periodic (or to be $\mathbb{R}$). In this case the domain is given by intervals given by $\sin(x)$ and it can be shifted only by $2k\pi$ to hold correctly. Therefore we can directly try with $T = 2\pi$:

$$f(x+2\pi) = \frac{\log(\sin(x+2\pi))}{\sin(x+2\pi) - 1} = \frac{\log(\sin(x))}{\sin(x) - 1} = f(x)$$

3) Lesson 23/10/26

3.1) Limits

Recall the following indeterminate forms:

$$(+\infty) + (-\infty) \tag{1}$$
$$0^{\pm} \cdot (\pm\infty) \tag{2}$$
$$\frac{0^{\pm}}{0^{\pm}} \tag{3}$$
$$\frac{\pm\infty}{\pm\infty} \tag{4}$$

Where D' is the
Limits at infinity are used to find asymptote. This has an equation of form:

$$y = mx + q$$

Two cases happen:

- $\lim_{x\to\infty} f(x) = c \in \mathbb{R} \longrightarrow$ asymptote is $y = c$ $(m = 0)$
- $\lim_{x\to\infty} f(x) = \pm\infty \longrightarrow$ here we might have an oblique asymptote $(m \neq 0)$ or a generic divergence. The slope can be found with:

  $$m = \lim_{x\to\pm\infty} \frac{f(x)}{x}$$

  if $m = \pm\infty$ there is no asymptote, otherwise q can be found as

  $$q = \lim f(x) - mx$$

1) Fraction

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = \frac{x-5}{x^2-9}$$

Domain:

$$D = x \in \mathbb{R} \setminus \{\pm 3\}$$

Sign:

- Numerator $> 0 \longrightarrow x > 5$
- Denominator $> 0 \longrightarrow x \in (-\infty, -3) \cup (3, +\infty)$

$$f(x) > 0 \iff x \in (-3, 3) \cup (5, +\infty)$$

Limits at infinity:

- Positive Infinity:

  $$\lim_{x\to\infty} f(x) = \frac{\infty}{\infty} \quad \text{indeterminate form !}$$

  Notice that:

$$f(x) = \frac{x(1-\frac{5}{x})}{x^2(1-\frac{9}{x^2})} = \frac{(1-\frac{5}{x})}{x(1-\frac{9}{x^2})}$$

The fractional terms tend to zero ($\lim_{x\to\infty} c/x^n = 0$) so

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{(1-\frac{5}{x})}{x(1-\frac{9}{x^2})} = \frac{1-0^+}{\infty(1-0^+)} = \frac{1^-}{\infty} = 0^+$$

- Negative Infinity:

$$\lim_{x\to-\infty} f(x) = \frac{1-0^-}{-\infty(1-0^-)} = \frac{1^+}{-\infty} = 0^-$$

Limits at accumulation points:
Here the use of the sign will be useful to determine if we have 0^+ or 0^- at the denominator

- At $+3$:

$$\lim_{x\to3^+} f(x) = \frac{-2}{0^+} = -\infty$$

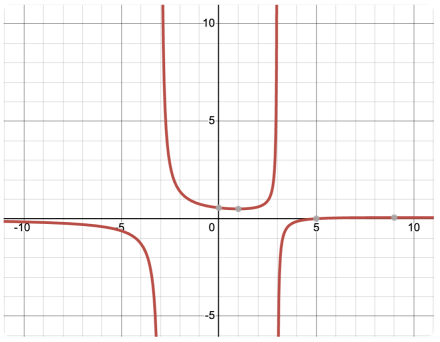
check also 3^-:

$$\lim_{x\to3^-} f(x) = \frac{-2}{0^-} = +\infty \neq \lim_{x\to3^+} f(x)$$

So the limit doesn't exist.

- At -3:

$$\lim_{x\to-3^+} f(x) = \frac{-8}{0^-} = +\infty \neq \lim_{x\to-3^-} f(x) = \frac{-8}{0^+} = -\infty$$



Graph

2) Oblique Asymptote

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = \frac{x^3}{x^2-1}$$

Domain:

$$D = x \in \mathbb{R} \setminus \{\pm1\}$$

Sign:

- Numerator $> 0 \longrightarrow x > 0$
- Denominator $> 0 \longrightarrow x \in (-\infty,-1) \cup (1,+\infty)$

$$f(x) > 0 \iff x \in (-1,0) \cup (1,+\infty)$$

Limits at infinity:

- Positive Infinity:

$$\lim_{x\to\infty} f(x) = \frac{\infty}{\infty} \quad \text{indeterminate form !}$$

Notice that:

$$f(x) = \frac{x^3}{x^2(1-\frac{1}{x^2})} = \frac{x}{1-\frac{1}{x^2}}$$

The fractional terms tend to zero ($\lim_{x\to\infty} c/x^n = 0$) so

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{x}{1-\frac{1}{x^2}} = \frac{\infty}{1-0^+} = \infty$$

Is there an oblique asymptote?

$$m = \lim_{x\to\infty} \frac{f(x)}{x} = \lim_{x\to\infty} \frac{1}{1-\frac{1}{x^2}} = 1$$
$$q = \lim_{x\to\infty} f(x) - x = \lim_{x\to\infty} \frac{1}{x(1-\frac{1}{x^2})} = 0^+$$

- Negative Infinity:

$$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} \frac{x}{1-\frac{1}{x^2}} = \frac{\infty}{1-0^+} = -\infty$$

Is there an oblique asymptote?

$$m = \lim_{x\to-\infty} \frac{f(x)}{x} = \lim_{x\to-\infty} \frac{1}{1-\frac{1}{x^2}} = 1$$
$$q = \lim_{x\to-\infty} f(x) - x = \lim_{x\to-\infty} \frac{1}{x(1-\frac{1}{x^2})} = 0^-$$

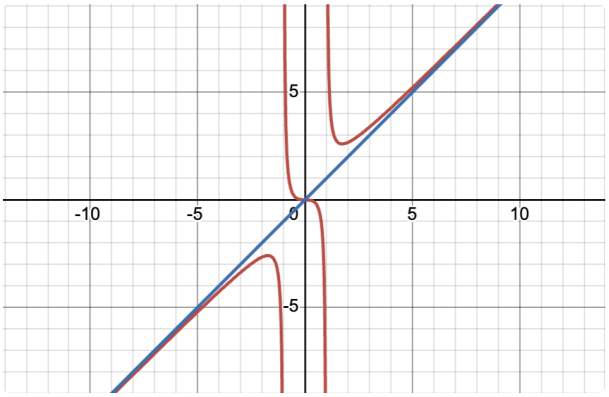
Limits at accumulation points:

- At $+1$:

$$\lim_{x\to1^+} f(x) = \frac{1}{0^+} = \infty \neq \lim_{x\to1^-} f(x) = \frac{1}{0^-} = -\infty$$

- At -1:

$$\lim_{x\to-1^+} f(x) = \frac{-1}{0^-} = +\infty \neq \lim_{x\to-1^-} f(x) = \frac{-1}{0^+} = -\infty$$



4) Log

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = \log\left(\frac{x+1}{x-1}\right)$$

Domain:

$$D = \left\{\frac{x+1}{x-1} > 0\right\} \cap \{x \neq 1\} = (-\infty, -1) \cup (1, +\infty)$$

Sign:

$$f(x) > 0 \iff \frac{x+1}{x-1} > 1 \to x > 1$$

We don't have any zero.

Limits at infinity:

- Positive Infinity:

  $$\lim_{x\to\infty} f(x) = \log\left(\frac{\infty}{\infty}\right) \quad \text{indeterminate form!}$$

  Notice that $\frac{x+1}{x-1} = \frac{x(1+1/x)}{x(1-1/x)} = \frac{1+\frac{1}{x}}{1-\frac{1}{x}}$ so

  $$\lim_{x\to\infty} f(x) = \log(1^+) = 0^+$$

  where we know that the solution must be positive from the domain.
  In general recall that a ratio of polynomial can be solved by just looking at the max degree of numerator and denominator!
- Negative Infinity:

  $$\lim_{x\to-\infty} f(x) = \log\left(1^-\right) = 0^-$$

  sign is deduced from previous sign analysis!

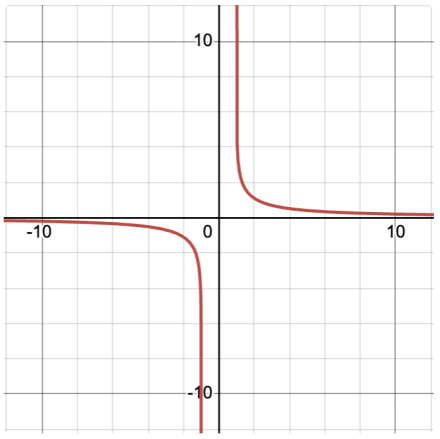
Limits at accumulation points:

- At 1 (only 1^+ due to domain):

  $$\lim_{x\to 1^+} f(x) = \log\left(\frac{2}{0^+}\right) = \log(+\infty) = \infty$$

- At -1 (only -1^- due to domain):

  $$\lim_{x\to -1^-} f(x) = \log\left(\frac{0^-}{-2}\right) = \log(0^+) = -\infty$$



Graph

7) Log o Log

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = \frac{\log(\log(x))}{1+\log(x)}$$

Domain:
We have three conditions:

- Inner logarithm requires: $x > 0$
- Outer logarithm in the numerator requires: $\log(x) > 0 \implies x > 1$
- Denominator must not vanish: $1 + \log(x) \neq 0 \implies \log(x) \neq -1 \implies x \neq \frac{1}{e}$
  Since $\frac{1}{e} \approx 0.37 < 1$, the condition $x \neq \frac{1}{e}$ is automatically satisfied for $x > 1$.

$$D = (1, +\infty)$$

Sign:

- Denominator: For all $x \in D = (1, +\infty)$, $\log(x) > 0$, so $1 + \log(x) > 1 > 0$ (always positive).
- Numerator:

$$\log(\log(x)) > 0 \iff \log(x) > 1 \iff x > e$$

$$f(x) > 0 \iff x \in (e, +\infty)$$

Limits at infinity:

- Positive Infinity:

$$\lim_{x \to +\infty} f(x) = \frac{+\infty}{+\infty} \quad \text{indeterminate form!}$$

  Substitute $t = \log(x)$.
  As $x \to +\infty$, $t \to +\infty$:

$$\lim_{t \to +\infty} \frac{\log(t)}{1+t} = \lim_{t \to +\infty} \frac{\log(t)}{t\left(1+\frac{1}{t}\right)} = \lim_{t \to +\infty} \frac{\log(t)}{t} \cdot \frac{1}{1+\frac{1}{t}}$$

  Recall the hierarchy of infinities ($\lim_{t \to +\infty} \frac{\log(t)}{t} = 0^+$):

$$\lim_{x \to +\infty} f(x) = 0^+ \cdot 1 = 0^+$$

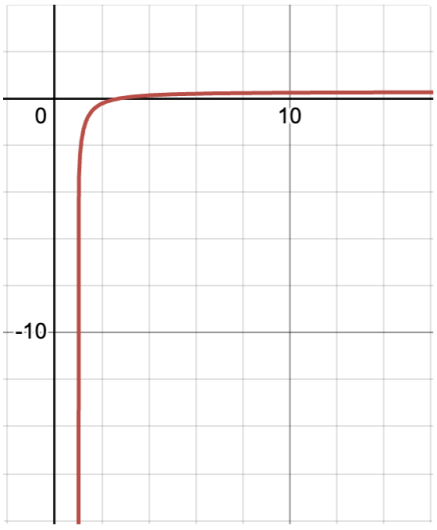
- Negative Infinity: Excluded by domain.

Limits at accumulation points:

- At 1 (only 1^+ due to domain):
  Numerator: as $x \to 1^+$, $\log(x) \to 0^+ \implies \log(\log(x)) \to -\infty$
  Denominator: $1 + \log(1^+) = 1 + 0^+ = 1^+$

$$\lim_{x \to 1^+} f(x) = \frac{-\infty}{1^+} = -\infty$$

  The line $x = 1$ is a vertical asymptote from the right.



Graph

8) Limited

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$$

Domain:

$$D = \{1 + x \geq 0\} \cap \{1 - x \geq 0\} \cap \{x \neq 0\} = [-1, 0) \cup (0, 1] = [-1, 1] \setminus \{0\}$$

Sign:
Two alternatives:

1. Find numerator denominator as usual (do at home):
   Numerator $> 0 \iff x \in (0, 1]$
   Denominator $> 0 \iff x > 0$
   so $f(x) > 0 \forall x \in D$
2. Rationalize

$$\begin{aligned} f(x) &= \frac{(\sqrt{1+x} - \sqrt{1-x})(\sqrt{1+x} + \sqrt{1-x})}{x(\sqrt{1+x} + \sqrt{1-x})} \\ &= \frac{(1+x) - (1-x)}{x(\sqrt{1+x} + \sqrt{1-x})} \\ &= \frac{2x}{x(\sqrt{1+x} + \sqrt{1-x})} \\ &= \frac{2}{\sqrt{1+x} + \sqrt{1-x}} \quad (\text{ only since } D \text{ excludes } 0) \end{aligned}$$

Since $\sqrt{1+x}+\sqrt{1-x}>0$ for all $x \in [-1,1]$:

$$f(x) > 0 \quad \forall x \in D$$

We don't have any zeros.

Limits at infinity:
Positive and Negative Infinity are excluded by domain ($D \subset [-1,1]$, so $\pm\infty$ are not accumulation points of D).
There are no horizontal or oblique asymptotes.

Limits at accumulation points:

- At 0:

$$\lim_{x\to 0} f(x) = \frac{0}{0} \quad \text{indeterminate form!}$$

Using the rationalized form:

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{2}{\sqrt{1+x}+\sqrt{1-x}} = \frac{2}{\sqrt{1}+\sqrt{1}} = \frac{2}{2} = 1$$

Notice that:

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(x) = 1$$

So the two-sided limit exists! $x=0$ is a removable discontinuity (eliminable singularity).

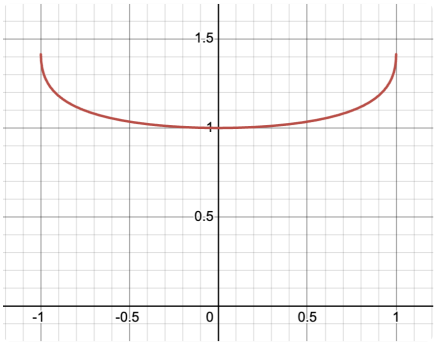
- At $+1$ (only 1^- due to domain): Since $1 \in D$, the function is continuous from the left:

$$\lim_{x\to 1^-} f(x) = f(1) = \frac{\sqrt{2}-0}{1} = \sqrt{2}$$

- At -1 (only $(-1)^+$ due to domain):Since $-1 \in D$, the function is continuous from the right:

$$\lim_{x\to -1^+} f(x) = f(-1) = \frac{0-\sqrt{2}}{-1} = \sqrt{2}$$

There are no vertical asymptotes.



Graph

4) Lesson 26/10/26

4.1) Standard Limits Exercises

3) Two Policemen (Harder!)

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = \log(e^{2x} - 1)$$

Domain:

$$D = \{e^{2x} - 1 > 0\} = \{x > 0\}$$

Sign:

$$f(x) > 0 \iff e^{2x} - 1 > 1 \to x > \frac{\log 2}{2}$$

Limits at infinity:

- Positive Infinity:

  $$\lim_{x\to\infty} f(x) = \log(\infty - 1) = \infty$$

  Is there an oblique asymptote?
  Notice how we can rewrite the logarithm:

  $$f(x) = \log\left[e^{2x}\left(1 - \frac{1}{e^{2x}}\right)\right] = \log\left(e^{2x}\right) + \log\left(1 - e^{-2x}\right) = 2x + \log\left(1 - e^{-2x}\right)$$

  so we have an oblique asymptote:

  $$m = \lim_{x\to\infty} \frac{f(x)}{x} = \lim_{x\to\infty} 2 + \frac{\log\left(1 - e^{-2x}\right)}{x} = 2 + \frac{\log(1^-)}{+\infty} = 2 + \frac{0^-}{+\infty} = 2$$

Alternatively the **squeeze theorem** can be used:

$$m = \lim_{x\to\infty} \frac{f(x)}{x} = \lim_{x\to\infty} \frac{\log(e^{2x} - 1)}{x}$$

Now find the bounds:

- Upper bound: $e^{2x} - 1 < e^{2x} \to \log(e^{2x} - 1) < 2x$
- Lower bound: $e^{2x} - 1 > \frac{e^{2x}}{2} \to \log(e^{2x} - 1) > 2x - \log(2)$ valid for large x ($x > 1$)
  Now divide by $x > 0$

  $$2 - \frac{\log(2)}{x} \leq \frac{\log(e^{2x} - 1)}{x} \leq 2$$

  Take the limit:

  $$2 \leq \lim_{x\to\infty} \frac{f(x)}{x} \leq 2$$

  The interception is:

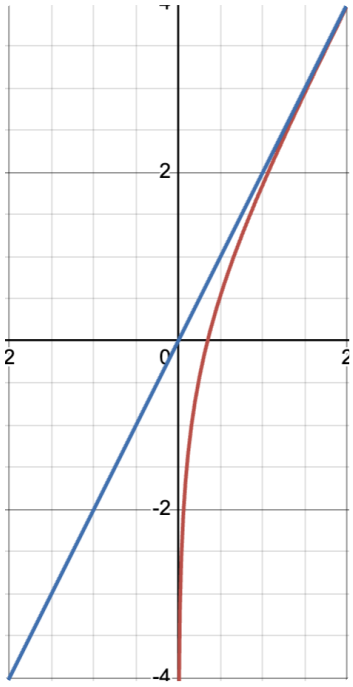
  $$\begin{aligned} q = \lim_{x\to\infty} f(x) - 2x &= +\infty - \infty \quad \text{indeterminate form!} \\ &= 2x + \log(1 - e^{-2x}) - 2x \\ &= \log(1 - e^{-2x}) = 0^- \end{aligned}$$

- Negative Infinity excluded by domain.

Limits at accumulation points:
Here we just need to check for 0^+ as 0^- is not in the domain!

- At 0^+:

$$\lim_{x \to 0^+} f(x) = \log(1^+ - 1) = \log(0^+) = -\infty$$



Asymptote in Blue

4.2) Taylor Expansion

A **Taylor expansion** takes a smooth function $f(x)$ near a specific point x_0 and replaces it with a **polynomial clone** $P_n(x)$ of degree n, plus a tiny error term (the remainder). This is useful as it transforms complicated functions in a simple polynomial function, which are easier to manipulate

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + o\Big((x - x_0)^n\Big) \quad \text{as } x \to x_0$$

But we will mostly limit our study cases at $x \to 0$ (for now)

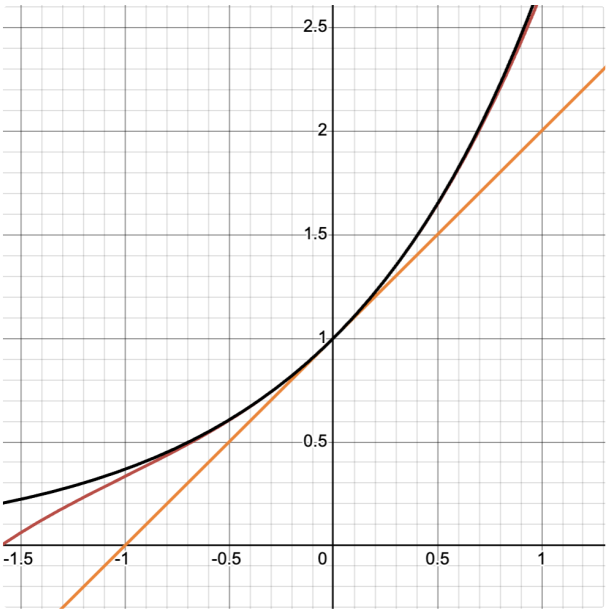
$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!}x^k + o(x^n) \quad \text{as } x \to 0$$

For now, since derivatives are not yet introduced, all the expansions will be provided.

Consider $f(x) = e^x$ at $x_0 = 0$. Its expansion is (up to order 3) $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$.
Now let's consider how well the various orders approximate $e^{0.5} = 1.64872$

Order n	$P_n(x)$	$P_n(0.5)$	$e^{0.5} - P_n(0.5)$
0	1	1	0.649
1	$1 + x$	1.5	0.149
2	$1 + x + \frac{1}{2}x^2$	1.625	0.024
3	$1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$	1.64583	0.0029

at order 3 it is already 99.8% accurate!



f(x) in black, 1st order expansion in orange, 3rd order expansion in red

4.2.1) Little-o Notation

Little o is the correction term. This encapsulates the difference of $f(x)$ and $P_n(x)$.

Definition 1 (Little-o)**.**

Given two functions $f(x)$ and $g(x)$ near x_0 (with $g(x) \neq 0$ in a punctured neighborhood of x_0), we say that $f(x)$ **is "little-o" of** $g(x)$ **as** $x \to x_0$, written $f(x) = o\Big(g(x)\Big)$, if:

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = 0$$

This is, in the Taylor expansion writing $o(x^n)$ means that all the next order terms go to zero faster than x^n and can be ignored.

For now let's just focus on $x \to 0$

Intuitively, little-o, refers to any function that shrinks faster then the specified function. Therefore we can say that $f(x) = o(g(x))$ is an abuse of notation as it should be $f(x) \in o(g(x))$ since $o(g(x))$ is a set that refers to all functions $h(x)$ where $h(x) \ll g(x)$

For example consider powers. As $x \to 0$ we have

$$x^n \ll x^m \quad \text{for } n > m$$

Therefore we can say:

- $x^3 = o(x^2)$ since $\lim_{x \to 0} x^3/x^2 = \lim_{x \to 0} x = 0$
- But not $x^2 = o(x^3)$ since $\lim_{x \to 0} x^2/x^3 = \lim_{x \to 0} 1/x = \pm\infty$

Theorem 2 (Little-o Properties)**.**

Here are some properties of little-o. Let $n > 0$*:*

- *Constant Multiplication:*

$$c \cdot o(x^n) = o(x^n) \text{ for any constant } c \neq 0 \qquad \text{The error has the same order}$$

- *Same Order Addition/subtraction:*

$$o(x^n) \pm o(x^n) = o(x^n) \quad \text{Important: } o(x^n) - o(x^n) \text{ is NOT } =0$$

- *Different order addition (Absorption):*

$$o(x^n) + o(x^m) = o(x^{\min(n,m)}) \quad \text{Ex: } o(x^2) + o(x^4) = o(x^2) \text{ since } x^4 \in o(x^2)$$

- *Absorbing Powers:*

$$c \cdot x^m + o(x^n) = o(x^n) \text{ if } m > n \quad \text{Ex: } 5x^7 + o(x^3) = o(x^3) \text{ since } x^7 \in o(x^3)$$

- *Multiplication by powers:*

$$x^m \cdot o(x^n) = o(x^{m+n}) \quad \text{Ex: } x^5 \cdot o(x^3) = o(x^8)$$

- *Product of two Little-o's:*

$$o(x^m) \cdot o(x^n) = o(x^{m+n}) \quad \text{Ex: } o(x^2) \cdot o(x^3) = o(x^5)$$

- *Powers of Little-o:*

$$[o(x^n)]^k = o(x^{n \cdot k}) \quad \text{Ex: } [o(x^2)]^3 = o(x^6)$$

Proofs (very quick):

1. Constant Multiplication: $c \cdot o(x^n) = o(x^n)$ $(c \neq 0)$
   Let $f(x) \in o(x^n)$, consider $g(x) = c \cdot f(x)$:

$$\lim_{x \to 0} \frac{g(x)}{x^n} = \lim_{x \to 0} \frac{c \cdot f(x)}{x^n} = c \cdot \lim_{x \to 0} \frac{f(x)}{x^n} = c \cdot 0 = 0$$

   Therefore, $c \cdot f(x) \in o(x^n)$.
2. Same-Order Addition / Subtraction: $o(x^n) \pm o(x^n) = o(x^n)$
   Let $f(x), g(x) \in o(x^n)$, consider $h(x) = f(x) \pm g(x)$:

$$\lim_{x \to 0} \frac{h(x)}{x^n} = \lim_{x \to 0} \left(\frac{f(x)}{x^n} \pm \frac{g(x)}{x^n} \right) = \lim_{x \to 0} \frac{f(x)}{x^n} \pm \lim_{x \to 0} \frac{g(x)}{x^n} = 0 \pm 0 = 0$$

   Hence, $f(x) \pm g(x) \in o(x^n)$.
3. Different-Order Addition (Absorption): $o(x^n) + o(x^m) = o(x^{\min(n,m)})$
   Assume (without loss of generality) that $n \leq m$, so $\min(n, m) = n$.
   Let $f(x) \in o(x^n)$ and $g(x) \in o(x^m)$.
   Examine the quotient with x^n:

$$\lim_{x \to 0} \frac{f(x) + g(x)}{x^n} = \lim_{x \to 0} \frac{f(x)}{x^n} + \lim_{x \to 0} \frac{g(x)}{x^n}$$

   By assumption, $\lim_{x \to 0} \frac{f(x)}{x^n} = 0$.
   For $g(x)$, rewrite the fraction by factoring out x^{m-n} (where $m - n \geq 0$):

$$\lim_{x \to 0} \frac{g(x)}{x^n} = \lim_{x \to 0} \left(\frac{g(x)}{x^m} \cdot x^{m-n} \right) = \left(\lim_{x \to 0} \frac{g(x)}{x^m} \right) \cdot \left(\lim_{x \to 0} x^{m-n} \right) = 0 \cdot \lim_{x \to 0} x^{m-n} = 0$$

   Thus:

$$\lim_{x\to 0}\frac{f(x)+g(x)}{x^n}=0+0=0 \implies f(x)+g(x)\in o(x^n)$$

4. Absorbing Powers: $c\cdot x^m+o(x^n)=o(x^n)$ if $m>n$
   Let $f(x)\in o(x^n)$ and $h(x)=cx^m+f(x)$ with $m>n$.
   Compute the limit:

$$\lim_{x\to 0}\frac{h(x)}{x^n}=\lim_{x\to 0}\left(\frac{cx^m}{x^n}+\frac{f(x)}{x^n}\right)=\lim_{x\to 0}\left(cx^{m-n}+\frac{f(x)}{x^n}\right)$$

   Because $m-n>0$:

$$\lim_{x\to 0}cx^{m-n}=0 \quad \text{and} \quad \lim_{x\to 0}\frac{f(x)}{x^n}=0$$

   Adding the limits yields $0+0=0$, confirming $cx^m+f(x)\in o(x^n)$.
5. Multiplication by Powers: $x^m\cdot o(x^n)=o(x^{m+n})$
   Let $f(x)\in o(x^n)$. Consider $g(x)=x^m\cdot f(x)$:

$$\lim_{x\to 0}\frac{g(x)}{x^{m+n}}=\lim_{x\to 0}\frac{x^m\cdot f(x)}{x^m\cdot x^n}=\lim_{x\to 0}\frac{f(x)}{x^n}=0$$

   Hence, $x^m\cdot f(x)\in o(x^{m+n})$.
6. Product of Two Little-o's: $o(x^m)\cdot o(x^n)=o(x^{m+n})$
   Let $f(x)\in o(x^m)$ and $g(x)\in o(x^n)$. Consider $h(x)=f(x)\cdot g(x)$:

$$\lim_{x\to 0}\frac{h(x)}{x^{m+n}}=\lim_{x\to 0}\left(\frac{f(x)}{x^m}\cdot\frac{g(x)}{x^n}\right)=\left(\lim_{x\to 0}\frac{f(x)}{x^m}\right)\cdot\left(\lim_{x\to 0}\frac{g(x)}{x^n}\right)=0\cdot 0=0$$

   Hence, $f(x)\cdot g(x)\in o(x^{m+n})$.
7. Powers of Little-o: $[o(x^n)]^k=o(x^{n\cdot k})$ ($k\in\mathbb{N}_{\geq 1}$)
   Let $f(x)\in o(x^n)$. Consider $g(x)=[f(x)]^k$:

$$\lim_{x\to 0}\frac{g(x)}{x^{n\cdot k}}=\lim_{x\to 0}\frac{[f(x)]^k}{(x^n)^k}=\lim_{x\to 0}\left(\frac{f(x)}{x^n}\right)^k$$

   By the continuous mapping theorem (or the product rule applied k times):

$$\lim_{x\to 0}\left(\frac{f(x)}{x^n}\right)^k=\left(\lim_{x\to 0}\frac{f(x)}{x^n}\right)^k=0^k=0$$

   Thus, $[f(x)]^k\in o(x^{n\cdot k})$.

4.3) Exercises

1) Little-o Warm Up Exercise:

Simplify the following expression as $x\to 0$:

$$E(x)=\left(x+2x^2+o(x^2)\right)-\left(x-x^2+4x^3+o(x^3)\right)$$

Do the basic algebraic simplifications:

$$E(x)=3x^2-4x^3+o(x^2)-o(x^3)$$

Since $o(x^2)+o(x^3)=o(x^2)$ we have

$$E(x)=3x^2-4x^3+o(x^2)$$

But notice that now $4x^3\in o(x^2)$ and thus

$$E(x)=3x^2+o(x^2)$$

2) Classic Cancellation Limit

Compute the limit:

$$\lim_{x\to 0}\frac{\sin(x)-x}{x^3}$$

Recall the expansion for sin:

$$\sin(x)=\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{(2n+1)!}=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots$$

This cannot be computed via the notable limit since:

$$\lim_{x\to 0}\frac{\sin(x)-x}{x^3}=\lim_{x\to 0}\frac{\sin(x)}{x^3}-\frac{1}{x^2}=\lim_{x\to 0}\frac{\sin(x)}{x}\cdot\frac{1}{x^2}-\frac{1}{x^2}$$

Where it is known that $\lim_{x\to 0}\sin(x)/x=1$ therefore:

$$\lim_{x\to 0}\frac{\sin(x)}{x}\cdot\frac{1}{x^2}-\frac{1}{x^2}=\lim_{x\to 0}\frac{1}{x^2}-\frac{1}{x^2}=0$$

But the last step is illegal, as the limit cannot be evaluated partially (that is we cannot simplify out $\sin(x)/x$ but keep $1/x^2$)

Try with order 1 expansion:

$$\lim_{x\to 0}\frac{\sin(x)-x}{x^3}=\lim_{x\to 0}\frac{x+o(x)-x}{x^3}=\lim_{x\to 0}\frac{o(x)}{x^3}$$

but since $o(x)/x^3=x^{-3}o(x)=o(x^{-2})$ we cannot determine it.

Since the denominator is of order 3, we expand until this order:

$$\lim_{x\to 0}\frac{x-\frac{x^3}{6}+o(x^3)-x}{x^3}=\lim_{x\to 0}\frac{-\frac{x^3}{6}+o(x^3)}{x^3}=\lim_{x\to 0}-\frac{1}{6}+\frac{o(x^3)}{x^3}=-\frac{1}{6}+0=-\frac{1}{6}$$

since $o(x^3)/x^3=x^{-3}o(x^3)=o(1)\to 0$ as $x\to 0$ and more in general $\lim_{x\to 0}o(x^n)/x^m=0$ if $n\geq m$.

3) Two Expansions:

Compute the limit:

$$\lim_{x\to 0}\frac{e^x-\cos(x)-x}{x^2}$$

Recall both expansions:

$$e^x=\sum_{n=0}^{\infty}\frac{x^n}{n!}=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots$$
$$\cos(x)=\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots$$

Apply them until order 2

$$\begin{aligned}e^x-\cos(x)-x&=\left(1+x+\frac{x^2}{2}+o(x^2)\right)-\left(1-\frac{x^2}{2}+o(x^2)\right)-x\\&=1-1+x-x+\frac{x^2}{2}+\frac{x^2}{2}+o(x^2)+o(x^2)\\&=x^2+o(x^2)\end{aligned}$$

Then the limit becomes:

$$\lim_{x\to 0}\frac{e^x-\cos(x)-x}{x^2}=\lim_{x\to 0}\frac{x^2+o(x^2)}{x^2}=1+0=1$$

4) Substitution

Find the Taylor expansion of order 4 at $x_0=0$ for $g(x)=e^{-x^2}$ and use it to compute:

$$\lim_{x\to 0}\frac{e^{-x^2}-\cos(x)}{x^4}$$

Recall both expansions:

$$e^x=\sum_{n=0}^{\infty}\frac{x^n}{n!}=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots$$
$$\cos(x)=\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots$$

We need to expand up to order 4. For cosine it is known, but e^{-x^2} doesn't have the basic formula.
Notice that with $t=-x^2$ we can expand e^t up to order 2 (since $o(t^2)\to o(x^4)$). This can be done since both tend to 0 as $x\to 0$

$$e^t=1+t+\frac{t^2}{2}+o(t^2)\to e^{-x^2}=1-x^2+\frac{x^4}{2}+o(x^4)$$

Now expand the limit:

$$\begin{aligned}\lim_{x\to 0}\frac{e^{-x^2}-\cos(x)}{x^4}&=\lim_{x\to 0}\frac{1-x^2+\frac{x^4}{2}+o(x^4)-1+\frac{x^2}{2}-\frac{x^4}{24}-o(x^4)}{x^4}\\&=\lim_{x\to 0}\frac{-\frac{1}{2}x^2+\frac{11}{24}x^4+o(x^4)}{x^4}\\&=\lim_{x\to 0}\frac{11}{24}-\frac{1}{2x^2}+\frac{o(x^4)}{x^4}\\&=\frac{11}{24}-\infty+0\\&=-\infty\end{aligned}$$

5) Exam 14.09.2020

Compute the limit:

$$\lim_{x\to 0^+}\frac{\log(1+\sinh x)-\sin x}{x^2}$$

Recall the expansions:

$$\sinh(x)=x+\frac{x^3}{6}+o(x^3)$$
$$\log(1+t)=t-\frac{t^2}{2}+o(t^2)$$
$$\sin(x)=x-\frac{x^3}{6}+o(x^3)$$

Here expansion of order two is needed. For the sin it's straight forward.
The log of sinh is done by substituting $t=\sinh(x)$, expanding the log and then the $\sinh(x)$.
How much do we need to expand?

$$\begin{aligned}\log(1+t) = t + o(t) &= \sinh(x) + o(\sinh(x)) \\ &= x + \frac{x^3}{6} + o(x^3) + o\left(x + \frac{x^3}{6} + o(x^3)\right) \\ &= x + \frac{x^3}{6} + o(x) \\ &= x + o(x)\end{aligned}$$

So not enough, in general if order n is needed, the expansion must be of order n

$$\begin{aligned}\log(1+t) = t - \frac{t^2}{2} + o(t^2) &= \sinh(x) - \frac{\sinh^2(x)}{2} + o(\sinh^2(x)) \\ &= x + o(x^2) - \frac{(x+o(x^2))^2}{2} + o([x+o(x^2)]^2) \\ &= x + o(x^2) - \frac{x^2 + 2xo(x^2) + o(x^2)^2}{2} + o(x^2 + o(x^3) + o(x^4)) \\ &= x - \frac{x^2}{2} + o(x^2)\end{aligned}$$

Now expand the whole limit:

$$\begin{aligned}\lim_{x\to 0^+} \frac{\log(1+\sinh x) - \sin x}{x^2} &= \lim_{x\to 0^+} \frac{x - \frac{x^2}{2} + o(x^2) - x + o(x^2)}{x^2} \\ &= \lim_{x\to 0^+} \frac{-\frac{x^2}{2} + o(x^2)}{x^2} \\ &= \lim_{x\to 0^+} -\frac{1}{2} + \frac{o(x^2)}{x^2} = -\frac{1}{2}\end{aligned}$$

5) Lesson 27/10/26

Warmup Exercise

Compute the limit:

$$\lim_{x\to 0} \frac{\log(\cos(x)) + \frac{x^2}{2}}{x^4}$$

Recall the expansions:

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$$
$$\log(1+t) = t - \frac{t^2}{2} + o(t^2)$$

Since the denominator is of order 4, we must expand the numerator up to $o(x^4)$.

Notice that $\log(\cos(x))$ is not a standard expansion, as the known formula is for $\log(1+t)$ as $t \to 0$. We cannot simply substitute $t = \cos(x)$ because $\cos(x) \to 1 \neq 0$ as $x \to 0$.

To fix this, we add and subtract 1 inside the logarithm, rewriting it as $\log\Big(1 + (\cos(x) - 1)\Big)$. Now the natural substitution is $t = \cos(x) - 1$, which is valid since $t \to 0$ as $x \to 0$.

Expanding $\cos(x)$ up to order 4 gives:

$$t = \cos(x) - 1 = -\frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$$

Notice that the lowest degree in t is x^2. Therefore, $\log(1+t)$ only needs to be expanded up to degree 2 in t (since $o(t^2) = o\Big((x^2)^2\Big) = o(x^4)$). Substituting t into $\log(1+t) = t - \frac{t^2}{2} + o(t^2)$, we get:

$$t = \cos(x) - 1 \quad \to \quad t = -\frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$$

Now notice that the lowest degree in t is x^2, so the $\log(1+t)$ just needs to be expanded to the second degree (since $o(x^2)^2 = o(x^4)$) and compute:

$$\begin{aligned}
\log(\cos(x)) = t - \frac{t^2}{2} + o(t^2) &= \left(-\frac{x^2}{2} + \frac{x^4}{24} + o(x^4)\right) - \frac{1}{2}\left(-\frac{x^2}{2} + \frac{x^4}{24} + o(x^4)\right)^2 + o(x^4) \\
&= -\frac{x^2}{2} + \frac{x^4}{24} - \frac{1}{2}\left(\frac{x^4}{4} + o(x^4)\right) + o(x^4) \\
&= -\frac{x^2}{2} + \frac{x^4}{24} - \frac{x^4}{8} + o(x^4) \\
&= -\frac{x^2}{2} - \frac{x^4}{12} + o(x^4)
\end{aligned}$$

Now replace the expansion into the whole limit:

$$\begin{aligned}
\lim_{x\to 0} \frac{\log(\cos(x)) + \frac{x^2}{2}}{x^4} &= \lim_{x\to 0} \frac{-\frac{x^2}{2} - \frac{x^4}{12} + o(x^4) + \frac{x^2}{2}}{x^4} \\
&= \lim_{x\to 0} \frac{-\frac{x^4}{12} + o(x^4)}{x^4} \\
&= \lim_{x\to 0} \left(-\frac{1}{12} + \frac{o(x^4)}{x^4}\right) \\
&= -\frac{1}{12} + 0 = -\frac{1}{12}
\end{aligned}$$

6) Exam 17.09.2019

Compute the limit:

$$\lim_{x\to 0^+} \frac{e^{x-2x^2}-1-x}{\sinh(x^2)+x^{7/3}\log x}$$

Recall the expansions:

$$\sinh(t) = t + \frac{t^3}{6} + o(t^3)$$
$$e^t = 1 + t + \frac{t^2}{2} + o(t^2)$$

Denominator:
First, look at $\sinh(x^2)$. Using $t = x^2$ (since $t \to 0$ as $x \to 0^+$), its leading term is of order 2:

$$\sinh(x^2) = \sinh(t) = t + o(t^2) = x^2 + o(x^4) = x^2 + o(x^2)$$

Next, look at $x^{7/3}\log(x)$. Since $\log(x)$ has no Taylor expansion at 0, we check whether this term is negligible compared to the leading x^2 term, that is, whether $x^{7/3}\log(x) \in o(x^2)$. Recall that $\lim_{x\to 0^+} x^\alpha \log x = 0$ for any $\alpha > 0$:

$$\lim_{x\to 0^+} \frac{x^{7/3}\log(x)}{x^2} = \lim_{x\to 0^+} x^{1/3}\log(x) = 0 \implies x^{7/3}\log(x) = o(x^2)$$

Combining both terms in the denominator and applying the absorption rule

$$\sinh(x^2) + x^{7/3}\log x = x^2 + o(x^4) + o(x^2) = x^2 + o(x^2)$$

Numerator:
Since the denominator is of order 2 and the $1 + x$ terms cancel out in the numerator, we expand e^t with $t = x - 2x^2$ up to order 2. Notice that $o(t^2) = o([x-2x^2]^2) = o(x^2)$, which absorbs all powers x^3 and higher:

$$\begin{aligned} e^{x-2x^2} = 1 + t + \frac{t^2}{2} + o(t^2) &= 1 + (x-2x^2) + \frac{(x-2x^2)^2}{2} + o([x-2x^2]^2) \\ &= 1 + x - 2x^2 + \frac{x^2 - 4x^3 + 4x^4}{2} + o(x^2) \\ &= 1 + x - 2x^2 + \frac{1}{2}x^2 + o(x^2) \\ &= 1 + x - \frac{3}{2}x^2 + o(x^2) \end{aligned}$$

Whole limit:

$$\begin{aligned} \lim_{x\to 0^+} \frac{e^{x-2x^2}-1-x}{\sinh(x^2)+x^{7/3}\log x} &= \lim_{x\to 0^+} \frac{1 + x - \frac{3}{2}x^2 + o(x^2) - 1 - x}{x^2 + o(x^2)} \\ &= \lim_{x\to 0^+} \frac{-\frac{3}{2}x^2 + o(x^2)}{x^2 + o(x^2)} \\ &= \lim_{x\to 0^+} \frac{x^2\left(-\frac{3}{2} + \frac{o(x^2)}{x^2}\right)}{x^2\left(1 + \frac{o(x^2)}{x^2}\right)} \\ &= \frac{-\frac{3}{2} + 0}{1 + 0} = -\frac{3}{2} \end{aligned}$$

7) Exam

Compute the limit:

$$\lim_{x\to 0}\frac{1-\cos(x^2)}{x(x-\sin x)+x^3\sin^2 x}$$

Recall the expansions:

$$\cos(t)=1-\frac{t^2}{2}+o(t^2)$$
$$\sin(x)=x-\frac{x^3}{6}+o(x^4)$$

Denominator:
If we only expand $\sin(x)$ to order 1, the first term in the denominator cancels out completely and leaves us with no leading term:

$$x(x-\sin x)=x\Big(x-(x+o(x))\Big)=x\cdot o(x)=o(x^2)$$

So we must expand $\sin(x)$ to the next non-zero term (order 3):

$$x(x-\sin x)=x\left(x-x+\frac{x^3}{6}+o(x^3)\right)=\frac{x^4}{6}+x\cdot o(x^3)=\frac{x^4}{6}+o(x^4)=\frac{x^4}{6}+o(x^4)$$

Now check the second term, $x^3\sin^2(x)$. We can see it is $o(x^4)$ either via the definition (using notable limit):

$$\lim_{x\to 0}\frac{x^3\sin^2(x)}{x^4}=\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^2\cdot x=1^2\cdot 0=0\implies x^3\sin^2(x)=o(x^4)$$

or directly via little-o algebra

$$\sin x=x+o(x)\implies x^3\sin^2 x=x^3(x+o(x))^2=x^5+o(x^5)=o(x^4)$$

Therefore, the denominator becomes:

$$x(x-\sin x)+x^3\sin^2 x=\frac{x^4}{6}+o(x^4)$$

Numerator:
Using $t=x^2$ in the cosine expansion up to order 2 in t (which gives order 4 in x):

$$1-\cos(x^2)=1-\left(1-\frac{(x^2)^2}{2}+o((x^2)^2)\right)=\frac{x^4}{2}+o(x^4)$$

Whole limit:

$$\begin{aligned}\lim_{x\to 0}\frac{1-\cos(x^2)}{x(x-\sin x)+x^3\sin^2 x}&=\lim_{x\to 0}\frac{\frac{x^4}{2}+o(x^4)}{\frac{x^4}{6}+o(x^4)}\\&=\lim_{x\to 0}\frac{x^4\left(\frac{1}{2}+\frac{o(x^4)}{x^4}\right)}{x^4\left(\frac{1}{6}+\frac{o(x^4)}{x^4}\right)}\\&=\frac{1/2}{1/6}=3\end{aligned}$$

8) Exam 12.09.2022

Compute the limit $\forall\alpha\in\mathbb{R}$:

$$\lim_{x\to 0^+} \frac{\sin x - \alpha x + \frac{1}{6}\alpha x^3}{\arctan(x^2+4x^3)}$$

First recall the standard expansions:

$$\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5)$$
$$\arctan(t) = t - \frac{t^3}{3} + \frac{t^5}{5} + o(t^5)$$

Numerator:
Since the polynomial terms go up to x^3, we expand $\sin(x)$ up to order 3:

$$\sin x - \alpha x + \frac{\alpha}{6}x^3 = x - \frac{x^3}{6} + o(x^3) - \alpha x + \frac{\alpha}{6}x^3 = (1-\alpha)x + \frac{\alpha - 1}{6}x^3 + o(x^3)$$

Denominator:
Use the substitution $t = x^2 + 4x^3$ (since $t \to 0$ as $x \to 0^+$) and expand $\arctan(t)$ to order 1.

$$\arctan(x^2+4x^3) = t + o(t) = x^2 + 4x^3 + o(x^2) = x^2 + o(x^2)$$

Now rewrite the entire limit:

$$\lim_{x\to 0^+} \frac{\sin x - \alpha x + \frac{1}{6}\alpha x^3}{\arctan(x^2+4x^3)} = \lim_{x\to 0^+} \frac{(1-\alpha)x + \frac{\alpha-1}{6}x^3 + o(x^3)}{x^2 + o(x^2)}$$

Now we split by cases depending on whether the leading terms in the numerator cancel out:

- **Case** $\alpha = 1$ (both the x and x^3 terms vanish):

$$\begin{aligned}\lim_{x\to 0^+} \frac{o(x^3)}{x^2 + o(x^2)} &= \lim_{x\to 0^+} \frac{o(x^3)}{x^2\left(1 + \frac{o(x^2)}{x^2}\right)} \\ &= \lim_{x\to 0^+} \frac{o(x^3)}{x^2} \cdot \frac{1}{1 + \frac{o(x^2)}{x^2}} \\ &= 0 \cdot \frac{1}{1+0} = 0\end{aligned}$$

  (since $\frac{o(x^3)}{x^2} = o(x) \to 0$ as $x \to 0^+$)
- **Case** $\alpha \neq 1$ (the order-1 term $(1-\alpha)x$ dominates the numerator):
  Factor out x from numerator and denominator:

$$\lim_{x\to 0^+} \frac{x\left[(1-\alpha) + \frac{\alpha-1}{6}x^2 + \frac{o(x^3)}{x}\right]}{x\left[x + \frac{o(x^2)}{x}\right]} = \lim_{x\to 0^+} \frac{1 - \alpha + \frac{\alpha-1}{6}x^2 + o(x^2)}{x + o(x)}$$
$$= \frac{1-\alpha}{0^+} = \begin{cases} +\infty & \text{if } \alpha < 1 \\ -\infty & \text{if } \alpha > 1 \end{cases}$$

So the final result is:

$$\lim_{x\to 0^+} \frac{\sin(x) - \alpha x + \frac{1}{6}\alpha x^3}{\arctan(x^2+4x^3)} = \begin{cases} +\infty & \text{if } \alpha < 1 \\ 0 & \text{if } \alpha = 1 \\ -\infty & \text{if } \alpha > 1 \end{cases}$$

In class however (see written notes) we aimed to have same order at numerator and denominator, however the result was the same if not with a bit more algebra.

6) Lesson 6/11/26

6.1) Standard Limit Exercise

5) Exponential (Harder)

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x) = e^{\frac{x^2-4}{x-1}}$$

Domain:

$$D = \{x - 1 \neq 0\} = \mathbb{R} \setminus \{1\} = (-\infty, 1) \cup (1, +\infty)$$

Sign:
Since the exponential function satisfies $e^t > 0$ for all $t \in \mathbb{R}$:

$$f(x) > 0 \quad \forall x \in D$$

We don't have any zeros.

Limits at infinity:

- Positive Infinity:

  $$\lim_{x \to +\infty} f(x) = e^{+\infty} = +\infty$$

  Is there an oblique asymptote?
  Check the slope m:

  $$m = \lim_{x \to +\infty} \frac{f(x)}{x} = \lim_{x \to +\infty} \frac{e^{\frac{x^2-4}{x-1}}}{x}$$

  Notice by polynomial division that $\frac{x^2-4}{x-1} = x + 1 - \frac{3}{x-1}$, so:

  $$\frac{f(x)}{x} = \frac{e^{x+1} \cdot e^{-\frac{3}{x-1}}}{x} = e \cdot e^{-\frac{3}{x-1}} \cdot \frac{e^x}{x}$$

  Therefore

  $$\lim_{x \to \infty} \frac{f(x)}{x} = e \cdot e^0 \cdot \infty = \infty$$

  There is no horizontal or oblique asymptote as $x \to +\infty$.
- Negative Infinity:

  $$\lim_{x \to -\infty} f(x) = e^{-\infty} = 0^+$$

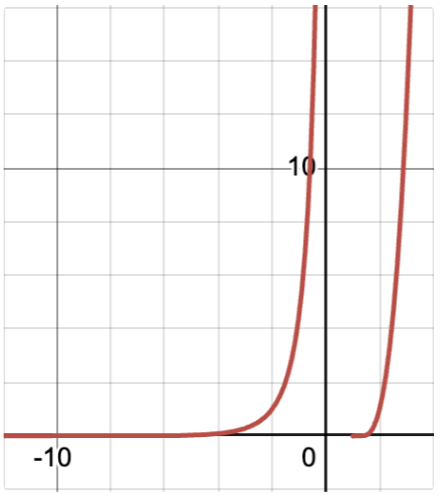
  We have a horizontal asymptote at $y = 0$ as $x \to -\infty$.

Limits at accumulation points:

- At 1

  $$\lim_{x \to 1^+} \frac{x^2 - 4}{x - 1} = \frac{-3}{0^+} = -\infty \implies \lim_{x \to 1^+} f(x) = e^{-\infty} = 0^+$$

  $$\lim_{x \to 1^-} \frac{x^2 - 4}{x - 1} = \frac{-3}{0^-} = +\infty \implies \lim_{x \to 1^-} f(x) = e^{+\infty} = +\infty$$



Graph

Polynomial division:
Add and Subtract (Difference of Squares Trick)
To make a multiple of the denominator $(x-1)$ appear in the numerator, recall that $x^2-1=(x-1)(x+1)$. We rewrite -4 as $-1-3$:

$$\begin{aligned}\frac{x^2-4}{x-1} &= \frac{(x^2-1)-3}{x-1} \\ &= \frac{x^2-1}{x-1} - \frac{3}{x-1} \\ &= \frac{(x-1)(x+1)}{x-1} - \frac{3}{x-1} \\ &= x+1-\frac{3}{x-1}\end{aligned}$$

6) Sqrt (Hardest)

Determine the natural domain, the sign, and the limits at infinity (if any) and at the accumulation points of the domain.

$$f(x)=5x+2-\sqrt{25x^2+12x}$$

Domain:
The roots of the argument of the sqrt are $x=-\frac{12}{25}$ and $x=0$. Taking the external intervals:

$$D=\left\{25x^2+12x\geq 0\right\}=\left(-\infty,-\frac{12}{25}\right]\cup[0,+\infty)$$

Sign:

$$f(x)>0 \iff \sqrt{25x^2+12x}<5x+2$$

As seen in first class (case $\sqrt{A(x)}<B(x)$) we have:

$$\begin{cases}A(x)\geq 0\\ B(x)>0\\ A(x)<[B(x)]^2\end{cases} \iff \begin{cases}x\in\left(-\infty,-\frac{12}{25}\right]\cup[0,+\infty)\\ 5x+2>0 \implies x>-\frac{2}{5}\\ 25x^2+12x<25x^2+20x+4 \implies x>-\frac{1}{2}\end{cases}$$

Then:

$$f(x)>0 \iff x\in[0,+\infty)$$

Limits at infinity:

- Positive Infinity:

$$\lim_{x\to+\infty} f(x) = (+\infty) - (+\infty) \quad \text{indeterminate form!}$$

Rationalize the expression:
To rationalize the idea is as follows:

$$a \pm \sqrt{b} = (a \pm \sqrt{b})\frac{a \mp \sqrt{b}}{a \mp \sqrt{b}} = \frac{a^2 - b}{a \mp \sqrt{b}}$$

Apply it directly

$$\begin{aligned} f(x) &= 2 + (\underbrace{5x}_{a} - \sqrt{\underbrace{25x^2 + 12x}_{b}}) \\ &= 2 + \frac{(5x - \sqrt{25x^2+12x})(5x + \sqrt{25x^2+12x})}{5x + \sqrt{25x^2+12x}} \\ &= 2 + \frac{25x^2 - (25x^2+12x)}{5x + \sqrt{25x^2+12x}} \qquad \$\$For \$x > 0\$, extract \$|x| = x\$: \$\$ \lim_{x\to+\infty} f(x) = \lim_{x\to+\infty} 2 \cdot \\ &= 2 - \frac{12x}{5x + \sqrt{25x^2+12x}} \\ &= 2 - \frac{12x}{5x + |x|\sqrt{25 + \frac{12}{x}}} \end{aligned}$$

- Negative Infinity:

$$\lim_{x\to-\infty} f(x) = 5(-\infty) + 2 - \sqrt{+\infty} = -\infty - \infty = -\infty$$

This is not indeterminate.
Is there an oblique asymptote?
Find the slope m (recall now since $x < 0 \to |x| = -x$):

$$m = \lim_{x\to-\infty} \frac{f(x)}{x} = \lim_{x\to-\infty} \left(5 + \frac{2}{x} - \frac{\sqrt{25x^2+12x}}{x}\right) = 5 - \lim_{x\to-\infty} \frac{-x\sqrt{25 + \frac{12}{x}}}{x} = 10$$

Now find the intercept q:

$$q = \lim_{x\to-\infty} [f(x) - 10x] = \lim_{x\to-\infty} [2 - 5x - \sqrt{25x^2+12x}] = 2 - \lim_{x\to-\infty} \left(5x + \sqrt{25x^2+12x}\right)$$

This is an indeterminate form $(-\infty) + (+\infty)$.
Rationalize

$$\begin{aligned} 5x + \sqrt{25x^2+12x} &= \left(5x + \sqrt{25x^2+12x}\right)\frac{5x - \sqrt{25x^2+12x}}{5x - \sqrt{25x^2+12x}} \\ &= \frac{25x^2 - \left(25x^2+12x\right)}{5x - \sqrt{25x^2+12x}} \\ &= \frac{-12x}{5x - \sqrt{25x^2+12x}} \\ &= \frac{-12x}{5x - |x|\sqrt{25 + \frac{12}{x}}} \end{aligned}$$

Then

$$q = 2 + \lim_{x\to-\infty} \frac{12x}{5x - |x|\sqrt{25 + \frac{12}{x}}} = 2 + \lim_{x\to-\infty} \frac{12}{5 + \sqrt{25 + \frac{12}{x}}} = 2 + \frac{12}{10} = \frac{16}{5}$$

We have an oblique asymptote:

$$y = 10x + \frac{16}{5} \quad \text{as } x \to -\infty$$

Limits at accumulation points:

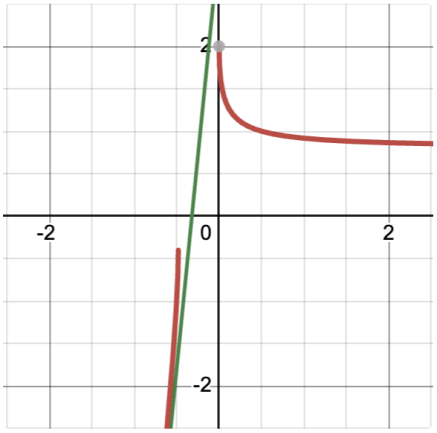
- At 0 (only 0^+ due to domain):

$$\lim_{x\to 0^+} f(x) = f(0) = 5(0) + 2 - 0 = 2$$

- At $-\frac{12}{25}$ (only $\left(-\frac{12}{25}\right)^-$ due to domain):

$$\lim_{x\to\left(-\frac{12}{25}\right)^-} f(x) = f\left(-\frac{12}{25}\right) = 5\left(-\frac{12}{25}\right) + 2 - 0 = -\frac{12}{5} + 2 = -\frac{2}{5}$$

There are no vertical asymptotes.



Asymptote in Green

6.2) Taylor Tips and Tricks

Substitution: To plug $t = f(x)$ into a standard Maclaurin expansion (centered at 0) as $x \to 0$, both x and t must tend to 0:

$$t = f(x) \text{ is valid iff } \lim_{x\to 0} f(x) = 0 \quad (\text{i.e., } t \to 0 \text{ as } x \to 0)$$

Lower Degree Change: Let $m, n \in \mathbb{R}$ such that $m < n$. As $x \to 0$:

$$o(x^n) \subset o(x^m)$$

Therefore, if needed, you can always absorb a higher-order remainder into a **lower** degree remainder, for example:

$$o(x^3) \to o(x^2) \quad \text{but NEVER} \quad o(x^2) \to o(x^3)$$

Polynomial and little-o inside little-o:
When you have a polynomial (or additional little-o terms) inside a little-o as $x \to 0$, the **lowest** non-zero degree always wins:

$$o\left(\sum_k a_k x^k\right) = o\left(x^{\min\{k\,:\,a_k\neq 0\}}\right)$$

For example:

$$o(x^2 + 3x^3 + o(x^4)) = o(x^2)$$

What degree? In general, the desired degree to expand to is the smallest one where the leading terms do not cancel out completely. If expanding $f(x) + g(x)$ to a low order n causes full cancellation:

$$f(x) + g(x) = o(x^n) \quad (\text{indeterminate if divided by } x^m)$$

you must expand to a higher order $m > n$ until a non-zero polynomial part $h(x) \neq 0$ survives:

$$f(x) + g(x) = h(x) + o(x^m)$$

In practice, first expand the simpler side (usually the denominator) until you find its first non-zero term $cx^n + o(x^n)$. Then aim to expand the numerator to at least that same order $o(x^n)$ so that, even if all terms in the numerator cancel out, you avoid indeterminate remainders of the form $\frac{o(x^m)}{x^n} = o(x^{m-n})$ with $m < n$.

6.3) Taylor Limit Exercises

Exercise

Compute the limit:

$$\lim_{x \to 0} \frac{(1 - \cosh(x))^2}{\sin(x)(x - \arctan(x)) + x^3 \sinh^2(x)}$$

Recall the expansions:

$$\begin{aligned}
\cosh(x) &= 1 + \frac{x^2}{2} + \frac{x^4}{24} + o(x^4) \\
\sin(x) &= x - \frac{x^3}{6} + o(x^3) \\
\arctan(x) &= x - \frac{x^3}{3} + o(x^3) \\
\sinh(x) &= x + \frac{x^3}{6} + o(x^3)
\end{aligned}$$

Numerator:
The numerator needs to be expanded at least to order two (otherwise the terms would cancel out completely)

$$1 - \cosh(x) = 1 - \left(1 + \frac{x^2}{2} + o(x^2)\right) = -\frac{x^2}{2} + o(x^2)$$

Now square the expression:

$$(1 - \cosh(x))^2 = \left(-\frac{x^2}{2} + o(x^2)\right)^2 = \frac{x^4}{4} - x^2 o(x^2) + [o(x^2)]^2 = \frac{x^4}{4} + o(x^4)$$

So the numerator is of order 4, denominator must be 4 or less!

Denominator:
Since the numerator is of order 4, we aim to expand the denominator up to $o(x^4)$.

Look at the first term, $\sin(x)(x - \arctan(x))$. Notice that expanding $\arctan(x)$ only to order 1 or 2 causes a cancellation inside the bracket:

$$x - \arctan(x) = x - (x + o(x^2)) = o(x^2)$$

So we must expand $\arctan(x)$ to order 3:

$$x - \arctan(x) = x - \left(x - \frac{x^3}{3} + o(x^3)\right) = \frac{x^3}{3} + o(x^3)$$

Since this bracket already starts at x^3, we only need to expand the outer factor $\sin(x)$ to order 1 ($\sin(x) = x + o(x)$) so that their product reaches order 4:

$$\begin{aligned}\sin(x)(x-\arctan(x)) &= (x+o(x))\left(\frac{x^3}{3}+o(x^3)\right)\\ &= \frac{x^4}{3}+x\cdot o(x^3)+\frac{x^3}{3}o(x)+o(x)o(x^3)\\ &= \frac{x^4}{3}+o(x^4)\end{aligned}$$

Now notice that the second term in the denominator, $x^3\sinh^2(x)$, is also $o(x^4)$ since:

$$\lim_{x\to 0}\frac{x^3\sinh^2(x)}{x^4}=\lim_{x\to 0}\left(\frac{\sinh(x)}{x}\right)^2\cdot x = 1^2\cdot 0 = 0 \implies x^3\sinh^2(x)=o(x^4)$$

(or directly via little-o algebra: $x^3(x+o(x))^2 = x^5+o(x^5)=o(x^4)$).

Therefore, the denominator becomes:

$$\sin(x)(x-\arctan(x))+x^3\sinh^2(x)=\frac{x^4}{3}+o(x^4)+o(x^4)=\frac{x^4}{3}+o(x^4)$$

Whole limit:

$$\begin{aligned}\lim_{x\to 0}\frac{(1-\cosh(x))^2}{\sin(x)(x-\arctan(x))+x^3\sinh^2(x)} &= \lim_{x\to 0}\frac{\frac{x^4}{4}+o(x^4)}{\frac{x^4}{3}+o(x^4)}\\ &= \lim_{x\to 0}\frac{x^4\left(\frac{1}{4}+\frac{o(x^4)}{x^4}\right)}{x^4\left(\frac{1}{3}+\frac{o(x^4)}{x^4}\right)}\\ &= \frac{1/4}{1/3}=\frac{3}{4}\end{aligned}$$

Parametric

Compute the limit $\forall\alpha>0$:

$$\lim_{x\to 0^+}\frac{\arctan(x)-\sin(x)+x^{10/3}\log(x)}{x^\alpha(1-\cos^2(x))}$$

First recall the standard expansions (as $x\to 0$) and standard logarithmic limit:

$$\begin{aligned}\sin(x) &= x-\frac{x^3}{6}+\frac{x^5}{120}+o(x^5)\\ \arctan(x) &= x-\frac{x^3}{3}+\frac{x^5}{5}+o(x^5)\\ \lim_{x\to 0^+}x^\beta\log(x) &= 0 \qquad \forall\beta>0\end{aligned}$$

Numerator:
Since the order-1 terms of $\arctan(x)$ and $\sin(x)$ cancel out ($x-x=0$), we must expand both functions up to order 3 (the first non-zero term):

$$\arctan(x)-\sin(x)=\left(x-\frac{x^3}{3}+o(x^3)\right)-\left(x-\frac{x^3}{6}+o(x^3)\right)=-\frac{1}{6}x^3+o(x^3)$$

Now let's compare the remaining term $x^{10/3}\log(x)$ with x^3. Since $\frac{10}{3}>3$ and polynomial decay dominates logarithmic growth as $x\to 0^+$, we can verify that $x^{10/3}\log(x)=o(x^3)$ by checking the definition of little-o:

$$\lim_{x\to 0^+}\frac{x^{10/3}\log(x)}{x^3}=\lim_{x\to 0^+}x^{1/3}\log(x)=0 \qquad\implies\qquad x^{10/3}\log(x)=o(x^3)$$

Putting the numerator together:

$$\arctan(x) - \sin(x) + x^{10/3}\log(x) = -\frac{1}{6}x^3 + o(x^3) + o(x^3) = -\frac{1}{6}x^3 + o(x^3)$$

Denominator:
Using the trigonometric identity $1 - \cos^2(x) = \sin^2(x)$ and expanding $\sin(x) = x + o(x)$ to order 1:

$$x^\alpha\Big(1 - \cos^2(x)\Big) = x^\alpha \sin^2(x) = x^\alpha\Big(x + o(x)\Big)^2 = x^\alpha\Big(x^2 + o(x^2)\Big) = x^{\alpha+2} + o(x^{\alpha+2})$$

(Note: You get the exact same result by expanding $\cos(x) = 1 - \frac{x^2}{2} + o(x^2)$ *and squaring it.)*

Now rewrite the entire limit:

$$\lim_{x\to 0^+} \frac{\arctan(x) - \sin(x) + x^{10/3}\log(x)}{x^\alpha(1 - \cos^2(x))} = \lim_{x\to 0^+} \frac{-\frac{1}{6}x^3 + o(x^3)}{x^{\alpha+2} + o(x^{\alpha+2})}$$

Factoring out the leading powers x^3 in the numerator and $x^{\alpha+2}$ in the denominator gives:

$$\lim_{x\to 0^+} \frac{x^3\left(-\frac{1}{6} + \frac{o(x^3)}{x^3}\right)}{x^{\alpha+2}\left(1 + \frac{o(x^{\alpha+2})}{x^{\alpha+2}}\right)} = \lim_{x\to 0^+} x^{1-\alpha} \cdot \frac{-\frac{1}{6} + \frac{o(x^3)}{x^3}}{1 + \frac{o(x^{\alpha+2})}{x^{\alpha+2}}}$$

Now we split by cases depending on the comparison between the numerator's order (3) and the denominator's order $(\alpha + 2)$:

- **Case** $\alpha = 1$ (numerator and denominator have the same order, $\alpha + 2 = 3$):
$$\lim_{x\to 0^+} x^0 \cdot \frac{-\frac{1}{6} + \frac{o(x^3)}{x^3}}{1 + \frac{o(x^3)}{x^3}} = \frac{-\frac{1}{6} + 0}{1 + 0} = -\frac{1}{6}$$
- **Case** $0 < \alpha < 1$ (the denominator has lower order $\alpha + 2 < 3$, so $1 - \alpha > 0$):
$$\lim_{x\to 0^+} x^{1-\alpha} \cdot \frac{-\frac{1}{6} + \frac{o(x^3)}{x^3}}{1 + \frac{o(x^{\alpha+2})}{x^{\alpha+2}}} = 0^+ \cdot \left(-\frac{1}{6}\right) = 0$$
- **Case** $\alpha > 1$ (the denominator has higher order $\alpha + 2 > 3$, so $\alpha - 1 > 0$):
$$\lim_{x\to 0^+} \frac{1}{x^{\alpha-1}} \cdot \frac{-\frac{1}{6} + \frac{o(x^3)}{x^3}}{1 + \frac{o(x^{\alpha+2})}{x^{\alpha+2}}} = \frac{-\frac{1}{6}}{0^+} = -\infty$$

So the final result is:

$$\lim_{x\to 0^+} \frac{\arctan(x) - \sin(x) + x^{10/3}\log(x)}{x^\alpha(1 - \cos^2(x))} = \begin{cases} 0 & \text{if } 0 < \alpha < 1 \\ -\frac{1}{6} & \text{if } \alpha = 1 \\ -\infty & \text{if } \alpha > 1 \end{cases}$$

Parametric (again)

Compute the limit $\forall \alpha \in \mathbb{R}$:

$$\lim_{x\to 0^+} \frac{\cosh(\alpha x) - e^{x^2} + x\log(\cos(x))}{x - \sin(x) + e^{-1/x^2}}$$

Recall the standard expansions:

$$\cosh(t) = 1 + \frac{t^2}{2} + \frac{t^4}{24} + o(t^4)$$
$$e^t = 1 + t + \frac{t^2}{2} + o(t^2)$$
$$\cos(x) = 1 - \frac{x^2}{2} + o(x^3)$$
$$\log(1+t) = t - \frac{t^2}{2} + o(t^2)$$
$$\sin(x) = x - \frac{x^3}{6} + o(x^3)$$

Denominator:
First expand $x - \sin(x)$. Since order 1 cancels out ($x - (x + o(x)) = o(x)$), we expand $\sin(x)$ up to order 3:

$$x - \sin(x) = x - \left(x - \frac{x^3}{6} + o(x^3)\right) = \frac{x^3}{6} + o(x^3)$$

Now notice that e^{-1/x^2} does not have a standard taylor expansion at 0 and the substitution technique fails since setting $t = 1/x^2$ we have that $x \to 0$ but $t \to \infty$. Again the definition must be used. For sake of exercises, instead of testing $o(x^3)$ we test the parametric limit $(o(x^n),\ n > 0)$:
Rewrite the fraction as a single exponential using $x^n = e^{n\log(x)}$ (for $x \to 0^+$):

$$\lim_{x\to 0^+} \frac{e^{-1/x^2}}{x^n} = \lim_{x\to 0^+} \frac{e^{-1/x^2}}{e^{n\log(x)}} = \lim_{x\to 0^+} e^{-\frac{1}{x^2} - n\log(x)}$$

Factor out $-\frac{1}{x^2}$ in the exponent:

$$\lim_{x\to 0^+} e^{-\frac{1}{x^2}\left(1 + nx^2\log(x)\right)} = e^{-\infty\cdot(1+0)} = e^{-\infty} = 0$$

(since $\lim_{x\to 0^+} x^2\log(x) = 0$ *by the standard hierarchy of infinities).*

Therefore we discovered that $e^{-1/x^2} = o(x^n),\ n > 0$ and we can freely chose $n = 3$ to comply with the previous expansion.
The denominator becomes therefore

$$\frac{x^3}{6} + o(x^3)$$

Numerator:
Since the denominator is of order 3, we aim to expand the numerator up to $o(x^3)$ (or stop earlier if a lower power survives):

1. For $\cosh(\alpha x)$, use $t = \alpha x$:

$$\cosh(t) = 1 + \frac{t^2}{2} + o(t^3) = 1 + \frac{(\alpha x)^2}{2} + o(\alpha^3 x^3) = 1 + \frac{\alpha^2}{2}x^2 + o(x^3)$$

   (since cosh *only has even powers, its order-3 term is* 0*).*
2. For e^{x^2}, use $t = x^2$ up to order 1 in t (since the next term is $\frac{t^2}{2} = \frac{x^4}{2} \in o(x^3)$):

$$e^t = 1 + t + \frac{t^2}{2} + o(t^2) = 1 + x^2 + \frac{x^4}{2} + o(x^4) = 1 + x^2 + o(x^3)$$

3. For $x\log(\cos(x))$, notice there is already an x in front, so we only need $\log(\cos(x))$ up to $o(x^2)$. Rewrite $\log(\cos(x)) = \log\left(1 + (\cos(x) - 1)\right)$ and use $t = \cos(x) - 1 = -\frac{x^2}{2} + o(x^2)$:

$$x\log(\cos(x)) = x\Big(t + o(t)\Big) = x\left(-\frac{x^2}{2} + o(x^2)\right) = -\frac{x^3}{2} + o(x^3)$$

Combining all three parts in the numerator:

$$\begin{aligned}\cosh(\alpha x) - e^{x^2} + x\log(\cos(x)) &= \left(1 + \frac{\alpha^2}{2}x^2 + o(x^3)\right) - \left(1 + x^2 + o(x^3)\right) - \frac{x^3}{2} + o(x^3) \\ &= \left(\frac{\alpha^2}{2} - 1\right)x^2 - \frac{1}{2}x^3 + o(x^3)\end{aligned}$$

Whole limit:

$$\lim_{x\to 0^+} \frac{\cosh(\alpha x) - e^{x^2} + x\log(\cos(x))}{x - \sin(x) + e^{-1/x^2}} = \lim_{x\to 0^+} \frac{\left(\frac{\alpha^2}{2} - 1\right)x^2 - \frac{1}{2}x^3 + o(x^3)}{\frac{1}{6}x^3 + o(x^3)}$$

Now we split by cases depending on whether the order-2 term in the numerator vanishes ($\frac{\alpha^2}{2} - 1 = 0 \iff \alpha = \pm\sqrt{2}$):

- **Case** $\alpha = \pm\sqrt{2}$ (the x^2 term cancels out, leaving order 3 on both top and bottom):

$$\begin{aligned}\lim_{x\to 0^+} \frac{-\frac{1}{2}x^3 + o(x^3)}{\frac{1}{6}x^3 + o(x^3)} &= \lim_{x\to 0^+} \frac{x^3\left(-\frac{1}{2} + \frac{o(x^3)}{x^3}\right)}{x^3\left(\frac{1}{6} + \frac{o(x^3)}{x^3}\right)} \\ &= \frac{-1/2}{1/6} = -3\end{aligned}$$

- **Case** $\alpha \neq \pm\sqrt{2}$ (the order-2 term dominates the numerator):
  Factor out x^2 from numerator and denominator:

$$\begin{aligned}\lim_{x\to 0^+} \frac{x^2\left[\left(\frac{\alpha^2}{2} - 1\right) - \frac{1}{2}x + \frac{o(x^3)}{x^2}\right]}{x^2\left[\frac{1}{6}x + \frac{o(x^3)}{x^2}\right]} &= \lim_{x\to 0^+} \frac{\left(\frac{\alpha^2}{2} - 1\right) - \frac{1}{2}x + o(x)}{\frac{1}{6}x + o(x)} \\ &= \frac{\frac{\alpha^2}{2} - 1}{0^+} = \begin{cases} +\infty & \text{if } \alpha < -\sqrt{2} \text{ or } \alpha > \sqrt{2} \quad \left(\frac{\alpha^2}{2} - 1 > 0\right) \\ -\infty & \text{if } -\sqrt{2} < \alpha < \sqrt{2} \quad \left(\frac{\alpha^2}{2} - 1 < 0\right) \end{cases}\end{aligned}$$

So the final result is:

$$\lim_{x\to 0^+} \frac{\cosh(\alpha x) - e^{x^2} + x\log(\cos(x))}{x - \sin(x) + e^{-1/x^2}} = \begin{cases} +\infty & \text{if } \alpha < -\sqrt{2} \text{ or } \alpha > \sqrt{2} \\ -3 & \text{if } \alpha = \pm\sqrt{2} \\ -\infty & \text{if } -\sqrt{2} < \alpha < \sqrt{2} \end{cases}$$

7) Lesson 13/11/26

7.1) Differentiability

Let's recap some theory

Definition (Differentiable on right/left)**.**

Let f be defined in a right/left neighborhood of $x_0 \in \mathbb{R}$. We say f is **differentiable on the right/left at** x_0 if the right/left limit of the difference quotient $\frac{\Delta f}{\Delta x}$ between x_0 and x exists finite for $x \to x_0$.
The real number

$$f'_{+/-}(x_0) = \lim_{x \to x_0^{+/-}} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{\Delta x \to 0^{+/-}} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

is the right/left derivative of f at x_0

Moreover we can say that a map is differentiable iff:

Theorem (Differentiability)**.**

A map f defined around a point $x_0 \in \mathbb{R}$ is differentiable at x_0 iff it is differentiable on both sides at x_0 and the left left and right derivatives coincide, in which case

$$f'(x_0) = f'_+(x_0) = f'_-(x_0)$$

Alternatively, if f is continuous at x_0 and differentiable at all points $x \neq x_0$ in a neighborhood of x_0. Then f is differentiable at x_0 provided that the limit of $f'(x)$ for $x \to x_0$ exists finite. If so

$$f'(x_0) = \lim_{x \to x_0} f'(x)$$

Where the continuity at x_0 must be verified

7.2) Exercises on Differentiability

b) Sin and abs value

Discuss differentiability at the point $x_0 = 0$:

$$f(x) = \sin|x|$$

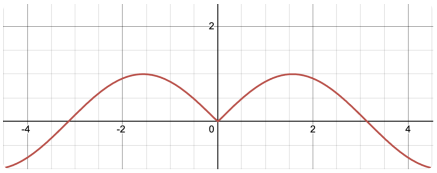
Using the definition we have

$$f'_+(0) = \lim_{x \to 0^+} \frac{\sin|x| - \sin(0)}{x - 0} = \lim_{x \to 0^+} \frac{\sin x}{x} = 1$$

However

$$f'_-(0) = \lim_{x \to 0^-} \frac{\sin|x| - \sin(0)}{x - 0} = \lim_{x \to 0^-} \frac{\sin -x}{x} = \lim_{x \to 0^-} \frac{-\sin x}{x} = -1$$

Since the limit is not the same, the function is not differentiable at $x_0 = 0$



Plot

c) Piecewise function

Discuss differentiability at the point $x_0 = 0$:

$$f(x) = \begin{cases} e^{-1/x^2} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Recall

$$\text{differentiable at } x_0 \implies \text{continuous at } x_0$$

which is equivalent to

$$\text{not continuous at } x_0 \implies \text{not differentiable at } x_0$$

A piecewise function is not necessarily continuous, let's check the continuity at x_0 first:

$$\lim_{x \to 0^+} e^{-1/x^2} = \lim_{x \to 0^-} e^{-1/x^2} = 0 = f(0)$$

Since it is continuous we must check the differentiability using the definition:

$$f'_+(0) = \lim_{x \to 0^+} \frac{e^{-1/x^2} - 0}{x - 0} = \lim_{x \to 0^+} \frac{e^{-1/x^2}}{x}$$

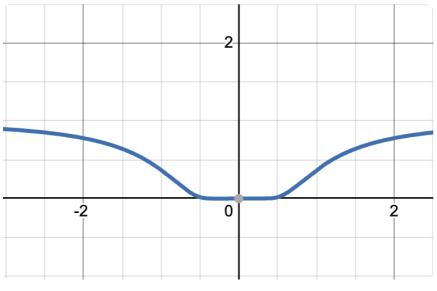
via substitution $t = \frac{1}{x}$ as $x \to 0^+$ and since exponential growth dominates linear growth

$$f'_+(0) = \lim_{t \to \infty} \frac{t}{e^{t^2}} = 0^+$$

Now the left limit:

$$f'_-(0) = \lim_{x \to 0^-} \frac{e^{-1/x^2} - 0}{x - 0} = \lim_{x \to 0^-} \frac{e^{-1/x^2}}{x} = \lim_{t \to -\infty} \frac{t}{e^{t^2}} = 0^-$$

Since both limits are equal the derivative at x_0 exists and is equal to 0.



Plot

d) Sqrt

Discuss differentiability at the point $x_0 = -1$:

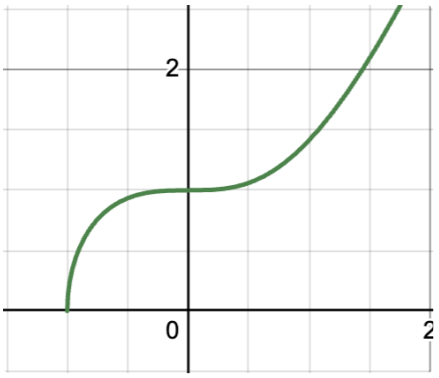
$$f(x) = \sqrt{1 + x^3}$$

Notice that the domain is $x \geq -1$ so we need to check only at -1^+

Using the definition we have

$$\begin{aligned}\lim_{x\to -1^+}\frac{\sqrt{1+x^3}-\sqrt{1-1}}{x-(-1)} &= \lim_{x\to -1^+}\frac{\sqrt{1+x^3}}{x+1}\\ &= \lim_{x\to -1^+}\frac{\sqrt{1+x^3}}{x+1}\frac{\sqrt{1+x^3}}{\sqrt{1+x^3}}\\ &= \lim_{x\to -1^+}\frac{1+x^3}{(x+1)\sqrt{1+x^3}}\\ &= \lim_{x\to -1^+}\frac{(x+1)(1-x+x^2)}{(x+1)\sqrt{1+x^3}}\\ &= \frac{3}{0^+} = +\infty\end{aligned}$$

Since the limit is not a finite real number, the function is not differentiable at $x_0 = -1$.



plot

7.3) Derivatives

Here is a list of the various derivatives of elementary functions

$f(x)$	$f'(x)$
x^α	$\alpha x^{\alpha-1}, \quad \forall \alpha \in \mathbb{R}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$1+\tan^2 x = \dfrac{1}{\cos^2 x}$
$\arcsin x$	$\dfrac{1}{\sqrt{1-x^2}}$
$\arccos x$	$-\dfrac{1}{\sqrt{1-x^2}}$
$\arctan x$	$\dfrac{1}{1+x^2}$
a^x	$(\log a)\, a^x$
$\log_a \lvert x\rvert$	$\dfrac{1}{(\log a)\, x}$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$

And some differentiation rules

$$\Big(\alpha f(x) + \beta g(x)\Big)' = \alpha f'(x) + \beta g'(x)$$
$$\Big(f(x)g(x)\Big)' = f'(x)g(x) + f(x)g'(x)$$
$$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{\Big(g(x)\Big)^2}$$
$$\Big(g(f(x))\Big)' = g'(f(x))f'(x)$$

When using a substitution the last differentiation rule (chain rule) must be applied

7.4) Exercises on derivatives

c) Piecewise function

Say where the following maps are differentiable and find the derivatives

$$f(x) = \begin{cases} x^2 + 1 & x \geq 0 \\ e^x - x & x < 0 \end{cases}$$

First verify the continuity:
Both parts of the definition are continuous, so we must just look at 0

$$\lim_{x \to 0^+} x^2 + 1 = \lim_{x \to 0^-} e^x - x = 1 = f(0)$$

Therefore it is continuous

Now compute the derivative for both definitions:

$$x^2 + 1 \overset{x^\alpha = \alpha x^{\alpha-1}}{\longrightarrow} 2x$$
$$e^x - x \overset{a^x = \log(a)a^x}{\longrightarrow} \log(e)e^x - 1 = e^x - 1$$

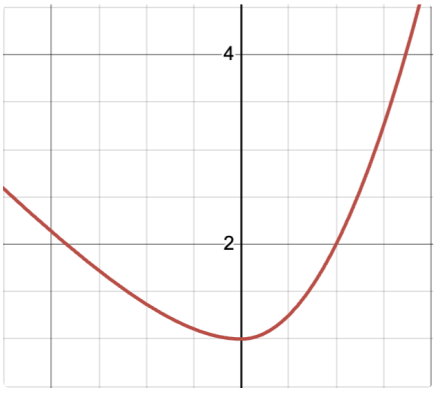
so we have

$$f'(x) = \begin{cases} 2x & x \geq 0 \\ e^x - 1 & x < 0 \end{cases}$$

Where the "$\geq$" is to be verified, as differentiability at 0 is not certain.
Both definitions are continuous, so differentiability needs only to be checked at $x_0 = 0$
Instead of using the definition, we use the theorem ($f'(x) = \lim_{x \to x_0} f'(x)$):

$$\lim_{x \to 0^+} 2x = \lim_{x \to 0^-} e^x - 1 = 0$$

So this function is fully differentiable.



Plot

a) Absolute value

Discuss where the following function is differentiable and find its derivative:

$$f(x) = x\sqrt{|x|}$$

The function is clearly continuous on all of $\mathbb{R}$, including at $x_0 = 0$.
One might suspect that since $|x|$ has a corner point at 0 (and $\sqrt{|x|}$ has a cusp), f might not be differentiable at $x_0 = 0$. Let's verify this.

Differentiate the function using the piecewise definition of absolute value:
First, rewrite $f(x)$ without the absolute value:

$$f(x) = x\sqrt{|x|} = \begin{cases} x\sqrt{x} & x \geq 0 \\ x\sqrt{-x} & x < 0 \end{cases} = \begin{cases} x^{3/2} & x \geq 0 \\ -(-x)^{3/2} & x < 0 \end{cases}$$

Now we differentiate both branches on the open intervals $x > 0$ and $x < 0$:

$$f'(x) = \begin{cases} \frac{3}{2}x^{1/2} & x > 0 \\ \frac{3}{2}(-x)^{1/2} & x < 0 \end{cases} = \begin{cases} \frac{3}{2}\sqrt{x} & x > 0 \\ \frac{3}{2}\sqrt{-x} & x < 0 \end{cases} = \frac{3}{2}\sqrt{|x|} \qquad \forall x \neq 0$$

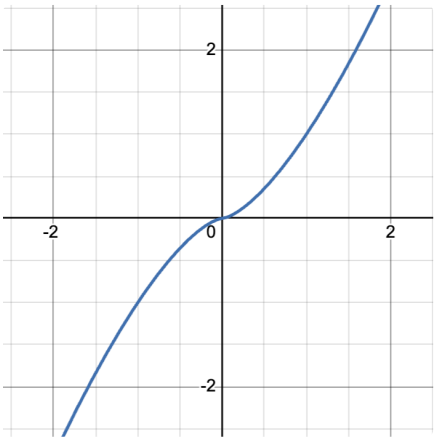
Here, the second branch ($x < 0$) was differentiated via the substitution $u(x) = -x$. Letting $\phi(u) = -u^{3/2}$, we are differentiating the composite function $\phi(u(x))$, which requires the chain rule:

$$\Big(\phi(u(x))\Big)' = \phi'(u(x)) \cdot u'(x) = -\frac{3}{2}u^{1/2} \cdot (-1) = \frac{3}{2}(-x)^{1/2}$$

Now we check differentiability at $x_0 = 0$. Since f is continuous at 0, we check the limits of $f'(x)$:

$$\lim_{x \to 0^+} \frac{3}{2}\sqrt{x} = \lim_{x \to 0^-} \frac{3}{2}\sqrt{-x} = 0$$

Thus f is differentiable at $x_0 = 0$ with $f'(0) = 0$! Our initial intuition was wrong (multiplying by x "smoothed out" the cusp at the origin), and $f'(x) = \frac{3}{2}\sqrt{|x|}$ holds for all $x \in \mathbb{R}$.



Plot

There is a second way (besides the definition) to calculate this limit:

First, as a preliminary exercise, consider the absolute value function:

$$a(x) = |x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

This is not differentiable at 0 since:

$$a'(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases} \quad \Longrightarrow \quad \lim_{x \to 0^+} a'(x) \neq \lim_{x \to 0^-} a'(x)$$

Notice that for $x \neq 0$, $a'(x)$ is precisely $\text{sign}(x) = \frac{|x|}{x}$.

Now we can compute $f'(x)$ for $x \neq 0$ using the product rule $\Big(h(x)g(x)\Big)' = h'(x)g(x) + h(x)g'(x)$ with $h(x) = x$ and $g(x) = \sqrt{|x|}$.
Since $g(x)$ is a composite function, we apply the chain rule $\Big(\psi(a(x))\Big)' = \psi'(a(x))a'(x)$ with $\psi(y) = \sqrt{y}$ and $a(x) = |x|$:

$$g'(x) = \frac{1}{2}|x|^{-1/2} \cdot \text{sign}(x) = \frac{1}{2\sqrt{|x|}} \frac{|x|}{x} = \frac{\sqrt{|x|}}{2x} \qquad \forall x \neq 0$$

Substituting this into the product rule for $x \neq 0$:

$$f'(x) = 1 \cdot \sqrt{|x|} + x \cdot \frac{\sqrt{|x|}}{2x} = \sqrt{|x|} + \frac{1}{2}\sqrt{|x|} = \frac{3}{2}\sqrt{|x|} \qquad \forall x \neq 0$$

Finally, since f is continuous at 0 and $\lim_{x \to 0} f'(x) = \lim_{x \to 0} \frac{3}{2}\sqrt{|x|} = 0$, we again conclude that $f'(0) = 0$, matching our first result.

Nested functions

Discuss where the following function is differentiable and find its derivative:

$$f(x) = \cos\left(e^{x^2+1}\right)$$

Since $f(x)$ is a double composition, we can break it down by introducing intermediate variables from the innermost function to the outermost:

- Innermost function: $u(x) = x^2 + 1 \overset{(x^\alpha)' = \alpha x^{\alpha-1}}{\longrightarrow} u'(x) = 2x$
- Middle function: $v(u) = e^u \overset{(a^u)' = (\log a)a^u}{\longrightarrow} v'(u) = (\log e)e^u = e^u$
- Outermost function: $g(v) = \cos v \overset{(\cos v)' = -\sin v}{\longrightarrow} g'(v) = -\sin v$

Now we apply the chain rule $\Big(g(v(x))\Big)' = g'(v(x)) \cdot v'(x)$ twice. First, for the outer layer:

$$f'(x) = g'\Big(v(u(x))\Big) \cdot \Big(v(u(x))\Big)' = -\sin\left(e^{x^2+1}\right) \cdot \left(e^{x^2+1}\right)'$$

Next, we apply the chain rule a second time to differentiate $v(u(x)) = e^{x^2+1}$:

$$\left(e^{x^2+1}\right)' = v'\Big(u(x)\Big) \cdot u'(x) = e^{x^2+1} \cdot (x^2+1)' = e^{x^2+1} \cdot 2x$$

Putting both pieces together, we obtain:

$$f'(x) = -\sin\left(e^{x^2+1}\right) \cdot e^{x^2+1} \cdot 2x = -2xe^{x^2+1}\sin\left(e^{x^2+1}\right) \qquad \forall x \in \mathbb{R}$$

Additionally one can compute it just in one run using the nested chain rule:
Once you are comfortable with the mechanism, you can apply the three-function chain rule $\Big(g(v(u(x)))\Big)' = g'(v(u(x))) \cdot v'(u(x)) \cdot u'(x)$ in a single line by differentiating from the outermost layer to the innermost:

$$
\begin{aligned}
f'(x) &= \left(\cos\left(e^{x^2+1}\right)\right)' \\
&= -\sin\left(e^{x^2+1}\right) \cdot \left(e^{x^2+1}\right)' \\
&= -\sin\left(e^{x^2+1}\right) \cdot e^{x^2+1} \cdot (x^2+1)' \\
&= -2xe^{x^2+1}\sin\left(e^{x^2+1}\right) \qquad \forall x \in \mathbb{R}
\end{aligned}
$$

General case (n nested functions)
By induction, let $f_1, f_2, \dots, f_n$ be functions such that the composite function

$$
F(x) = (f_n \circ f_{n-1} \circ \cdots \circ f_2 \circ f_1)(x) = f_n\Big(f_{n-1}\Big(\dots f_2(f_1(x))\dots\Big)\Big)
$$

is well-defined. If f_1 is differentiable at x, f_2 is differentiable at $f_1(x)$, …, and f_n is differentiable at $f_{n-1}(\dots f_1(x)\dots)$, then F is differentiable at x and its derivative is the product of the derivatives of each layer, evaluated at everything inside it ("peeling the onion" from the outside in):

$$
\begin{aligned}
F'(x) &= f_n'\Big(f_{n-1}\Big(\dots f_1(x)\dots\Big)\Big) \cdot f_{n-1}'\Big(f_{n-2}\Big(\dots f_1(x)\dots\Big)\Big) \cdots f_2'\Big(f_1(x)\Big) \cdot f_1'(x) \\
&= f_1'(x) \cdot \prod_{k=2}^{n} f_k'\Big((f_{k-1} \circ \cdots \circ f_1)(x)\Big)
\end{aligned}
$$

7.5) Exercises on past exams

Exam 07.02.2022

Given the function

$$
f(x) = \log(|x| - x^2 + x)
$$

i) determine the domain, symmetry, sign and compute limits and asymptotes at extremes of domain
ii) study derivability and find the first derivative

i) **Domain**
First, rewrite the argument of the logarithm without the absolute value using its piecewise definition:

$$
|x| - x^2 + x = \begin{cases} x - x^2 + x & x \geq 0 \\ -x - x^2 + x & x < 0 \end{cases} = \begin{cases} 2x - x^2 & x \geq 0 \\ -x^2 & x < 0 \end{cases}
$$

For $f(x)$ to be well-defined, the argument of the logarithm must be strictly positive. We check both branches:

- For $x < 0$, we need $-x^2 > 0 \iff x^2 < 0$, which is never satisfied in $\mathbb{R}$.
- For $x \geq 0$, we need $2x - x^2 > 0 \iff x(2-x) > 0$, which holds for $0 < x < 2$.

Thus, the domain is $\mathrm{Dom}(f) = (0, 2)$. Notice that since the entire domain lies in $(0, +\infty)$, the negative branch disappears completely and we can simply write:

$$
f(x) = \log(2x - x^2) \qquad \forall x \in (0, 2)
$$

Since $\mathrm{Dom}(f) = (0, 2)$ is not symmetric with respect to the origin, f cannot be even or odd.

Sign
Now we study the sign of $f(x)$ on $(0, 2)$:

$$
f(x) \geq 0 \iff \log(2x - x^2) \geq 0 \iff 2x - x^2 \geq 1
$$

Rearranging the terms gives $x^2 - 2x + 1 \leq 0 \iff (x-1)^2 \leq 0$. Since a square is never negative, this holds only when $(x-1)^2 = 0$, that is at $x = 1$. Thus, $f(x) = 0$ at $x = 1$, is strictly negative for all $x \in (0, 1) \cup (1, 2)$, and is never positive.

Limits
Next, we compute the limits at the extremes of the domain, 0^+ and 2^-:

$$\lim_{x\to 0^+} \log(2x - x^2) = \log(0^+) = -\infty$$

$$\lim_{x\to 2^-} \log(2x - x^2) = \log(0^+) = -\infty$$

Therefore, f has two vertical asymptotes: the lines $x = 0$ (as $x \to 0^+$) and $x = 2$ (as $x \to 2^-$). Since the domain is bounded, it makes no sense to take limits as $x \to \pm\infty$, so there are no horizontal or oblique asymptotes.

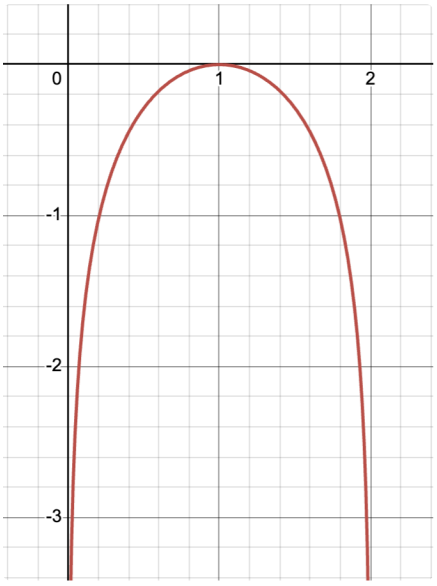
ii) **Differentiability and first derivative**

One might suspect that since $|x|$ has a corner point at $x_0 = 0$, we need to be careful with differentiability at the origin. However, as we just found, $x_0 = 0 \notin \mathrm{Dom}(f)$!

On its entire domain $(0, 2)$, the function reduces to $f(x) = \log(2x - x^2)$, which is the composition of the polynomial $u(x) = 2x - x^2$ (differentiable on all of $\mathbb{R}$ and strictly positive on $(0, 2)$) with $\phi(u) = \log(u)$ (differentiable for $u > 0$). By the chain rule, f is differentiable on all of $(0, 2)$, so $\mathrm{Dom}(f') = (0, 2)$.

Differentiating the composite function $\phi(u(x))$ gives:

$$f'(x) = \Big(\phi(u(x))\Big)' = \phi'(u(x)) \cdot u'(x) = \frac{1}{2x - x^2} \cdot (2 - 2x) = \frac{2(1 - x)}{x(2 - x)} \qquad \forall x \in (0, 2)$$



Plot

Exam

Given the function

$$f(x) = |x - 2|e^{1/(x-2)^2}$$

i) determine the domain, symmetry, sign and compute limits and asymptotes at extremes of domain
ii) study derivability and find the first derivative

i) Domain, symmetry, sign, limits and asymptotes

Domain
The exponential function and the absolute value are defined everywhere on $\mathbb{R}$, so the only restriction comes from the denominator in the exponent, $(x - 2)^2 \neq 0$, which requires $x \neq 2$. Thus, the domain is $\mathrm{Dom}(f) = \mathbb{R} \setminus \{2\} = (-\infty, 2) \cup (2, +\infty)$.

Symmetries
Since $\mathrm{Dom}(f) = \mathbb{R} \setminus \{2\}$ is not symmetric with respect to the origin ($-2 \in \mathrm{Dom}(f)$ while $2 \notin \mathrm{Dom}(f)$), f is neither even nor odd. (However, since every x appears in the form $(x-2)^2$ or $|x-2|$, the function is symmetric with respect to the vertical line $x = 2$, i.e., $f(2+t) = f(2-t)$ for all $t \neq 0$).

Sign
Now we study the sign of $f(x)$ on $\mathbb{R} \setminus \{2\}$. Since the exponential $e^{1/(x-2)^2}$ is strictly positive for all $x \neq 2$ and $|x-2| > 0$ for all $x \neq 2$, the product of two strictly positive quantities is strictly positive:

$$f(x) > 0 \qquad \forall x \in \mathbb{R} \setminus \{2\}$$

In particular, f has no zeros and is never negative.

Limits
Now we check the behavior at $\pm\infty$:

$$\lim_{x \to \pm\infty} |x-2| e^{1/(x-2)^2} = (+\infty) \cdot e^0 = +\infty$$

Thus, there are no horizontal asymptotes. Since the limit is infinite, we check for oblique asymptotes $y = mx + q$ as $x \to +\infty$ and $x \to -\infty$:

- As $x \to +\infty$, $|x-2| = x - 2 > 0$:

$$m_+ = \lim_{x \to +\infty} \frac{f(x)}{x} = \lim_{x \to +\infty} \frac{x-2}{x} e^{1/(x-2)^2} = 1 \cdot e^0 = 1$$

$$q_+ = \lim_{x \to +\infty} \Big(f(x) - x \Big) = \lim_{x \to +\infty} \Big[(x-2) e^{1/(x-2)^2} - (x-2) - 2 \Big] = \lim_{x \to +\infty} (x-2) \left(e^{1/(x-2)^2} - 1 \right) - 2$$

Now do a change of variable $y = x - 2$ and $x \to +\infty$, $y \to +\infty$, and $x = y + 2$

$$\begin{aligned} q_+ &= \lim_{y \to +\infty} \Big(y e^{1/y^2} - (y+2) \Big) \\ &= \lim_{y \to +\infty} \Big[y \left(e^{1/y^2} - 1 \right) - 2 \Big] \\ &= \lim_{y \to +\infty} \left[y \cdot \frac{1}{y^2} \cdot \frac{e^{1/y^2} - 1}{1/y^2} - 2 \right] \\ &= \lim_{y \to +\infty} \left[\frac{1}{y} \cdot \frac{e^{1/y^2} - 1}{1/y^2} - 2 \right] = 0 \cdot 1 - 2 = -2 \end{aligned}$$

Where the last step uses the notable limit

$$\lim_{t \to 0} \frac{e^t - 1}{t} = 1$$

Thus, $y = x - 2$ is a right oblique asymptote as $x \to +\infty$.

- As $x \to -\infty$, $|x-2| = -(x-2) > 0$:

$$m_- = \lim_{x \to -\infty} \frac{f(x)}{x} = \lim_{x \to -\infty} \frac{-(x-2)}{x} e^{1/(x-2)^2} = -1 \cdot e^0 = -1$$

$$q_- = \lim_{x \to -\infty} \Big(f(x) - (-x) \Big) = \lim_{x \to -\infty} \Big[-(x-2) e^{1/(x-2)^2} + x \Big]$$

Now do the same change of variable $y = x - 2$ with $x \to -\infty$, $y \to -\infty$, and $x = y + 2$:

$$
\begin{aligned}
q_- &= \lim_{y\to-\infty}\left(-ye^{1/y^2}+(y+2)\right)\\
&= \lim_{y\to-\infty}\left[-y\left(e^{1/y^2}-1\right)+2\right]\\
&= \lim_{y\to-\infty}\left[-y\cdot\frac{1}{y^2}\cdot\frac{e^{1/y^2}-1}{1/y^2}+2\right]\\
&= \lim_{y\to-\infty}\left[-\frac{1}{y}\cdot\frac{e^{1/y^2}-1}{1/y^2}+2\right] = 0\cdot 1+2 = 2
\end{aligned}
$$

Where the last step uses the notable limit $\lim_{t\to0}\frac{e^t-1}{t}=1$.
Thus, $y=-x+2$ is a left oblique asymptote as $x\to-\infty$.

Next, we check the behavior at $x\to2^{\pm}$:
As $x\to2^{\pm}$, $|x-2|\to0^+$ and $e^{1/(x-2)^2}\to e^{+\infty}=+\infty$, giving an indeterminate form $0\cdot\infty$.
Doing the change of variable $t=\frac{1}{|x-2|}$, as $x\to2^{\pm}$ we have $t\to+\infty$ and $(x-2)^2=|x-2|^2=\frac{1}{t^2}$:

$$
\lim_{x\to2^{\pm}}|x-2|e^{1/(x-2)^2}=\lim_{t\to+\infty}\frac{e^{t^2}}{t}=+\infty
$$

by the standard hierarchy of infinities. Therefore, the line $x=2$ is a bilateral vertical asymptote.

ii) Differentiability and first derivative
One might suspect that $|x-2|$ introduces a corner point at $x_0=2$. However, as we already established, $x_0=2\notin\mathrm{Dom}(f)$, so we don't even have to check differentiability at $x_0=2$!

First, rewrite $f(x)$ on $\mathbb{R}\setminus\{2\}$ without the absolute value:

$$
f(x)=|x-2|e^{1/(x-2)^2}=\begin{cases}(x-2)e^{(x-2)^{-2}} & x>2\\ -(x-2)e^{(x-2)^{-2}} & x<2\end{cases}
$$

On both open intervals $(-\infty,2)$ and $(2,+\infty)$, f is the product and composition of differentiable functions, so f is differentiable on its entire domain: $\mathrm{Dom}(f')=\mathbb{R}\setminus\{2\}$.

First, let's differentiate the exponential term $e^{(x-2)^{-2}}$ via the chain rule. Letting $u(x)=(x-2)^{-2}$ and $\phi(u)=e^u$, we have:

$$
\left(e^{(x-2)^{-2}}\right)'=e^{(x-2)^{-2}}\cdot\left((x-2)^{-2}\right)'=e^{1/(x-2)^2}\cdot\left(-2(x-2)^{-3}\right)=-\frac{2}{(x-2)^3}e^{1/(x-2)^2}
$$

Now we differentiate the first branch ($x>2$) using the product rule:

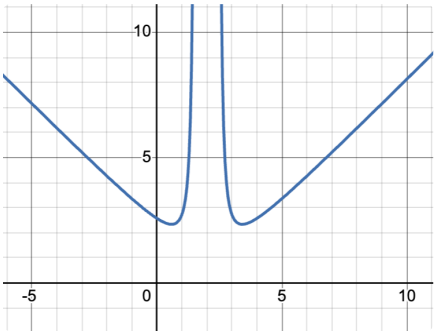
$$
f'(x)=1\cdot e^{1/(x-2)^2}+(x-2)\cdot\left(-\frac{2}{(x-2)^3}e^{1/(x-2)^2}\right)=e^{1/(x-2)^2}\left(1-\frac{2}{(x-2)^2}\right)=e^{1/(x-2)^2}\frac{(x-2)^2-2}{(x-2)^2}
$$

For the second branch ($x<2$), since $f(x)$ differs only by a minus sign in front of $(x-2)$, we immediately obtain:

$$
f'(x)=-e^{1/(x-2)^2}\frac{(x-2)^2-2}{(x-2)^2}
$$

Combining both branches into a single expression (or recalling that $\left(|x-2|\right)'=\frac{|x-2|}{x-2}=\mathrm{sgn}(x-2)$ for $x\neq2$), we can write:

$$f'(x) = \begin{cases} e^{1/(x-2)^2}\dfrac{(x-2)^2-2}{(x-2)^2} & x > 2 \\ -e^{1/(x-2)^2}\dfrac{(x-2)^2-2}{(x-2)^2} & x < 2 \end{cases} = \frac{|x-2|}{x-2}\cdot\frac{(x-2)^2-2}{(x-2)^2}e^{1/(x-2)^2} \qquad \forall x \neq 2$$



Plot

Bonus exercise: Compute the limits of the derivative

Now we compute the limits of $f'(x)$ at $\pm\infty$ and $2^{\pm}$. First, rewrite the rational factor as:

$$\frac{(x-2)^2-2}{(x-2)^2} = 1 - \frac{2}{(x-2)^2}$$

- At $\pm\infty$: as $x \to \pm\infty$, we have $\frac{1}{(x-2)^2} \to 0$, so $e^{1/(x-2)^2} \to e^0 = 1$ and $1 - \frac{2}{(x-2)^2} \to 1$. Using the respective branches:

$$\lim_{x\to+\infty} f'(x) = \lim_{x\to+\infty} e^{1/(x-2)^2}\left(1 - \frac{2}{(x-2)^2}\right) = 1 \cdot 1 = 1$$

$$\lim_{x\to-\infty} f'(x) = \lim_{x\to-\infty} -e^{1/(x-2)^2}\left(1 - \frac{2}{(x-2)^2}\right) = -1 \cdot 1 = -1$$

  (Notice that these coincide with the slopes $m_+ = 1$ and $m_- = -1$ of the oblique asymptotes).
- At $2^{\pm}$: as $x \to 2^{\pm}$, we have $(x-2)^2 \to 0^+$, so $1 - \frac{2}{(x-2)^2} \to -\infty$ and $e^{1/(x-2)^2} \to +\infty$ (which is not an indeterminate form!):

$$\lim_{x\to2^+} f'(x) = \lim_{x\to2^+} e^{1/(x-2)^2}\left(1 - \frac{2}{(x-2)^2}\right) = (+\infty)\cdot(-\infty) = -\infty$$

$$\lim_{x\to2^-} f'(x) = \lim_{x\to2^-} -e^{1/(x-2)^2}\left(1 - \frac{2}{(x-2)^2}\right) = -(+\infty)\cdot(-\infty) = +\infty$$

8) Symbols

- $\implies$: implies, sufficient condition ($P \implies Q$ means P is sufficient for Q, Q is necessary for P).
- $\impliedby$: implied by, necessary condition
- $\iff$: necessary + sufficient, if and only if "iff"
- $\emptyset$: empty set
- $\mathbb{N}$: natural numbers; $\{0, 1, 2, ...\}$
- $\mathbb{Z}$: integer numbers; $\{..., -2, -1, 0, 1, 2, ...\}$
- $\mathbb{Q}$: rational numbers; $\left\{ \frac{m}{n} \; m, n \in \mathbb{Z}, n \neq 0 \right\}$
- $\mathbb{R}$: real numbers
- $\mathbb{C}$: complex numbers
- $\cup$: Union; or
- $\cap$: Intersection; and
- $\exists$: there exists

9) Taylor Expansions

Here is a list of some of the most frequent expansions

$$e^x = 1 + x + \frac{x^2}{2} + \cdots + \frac{x^k}{k!} + \cdots + \frac{x^n}{n!} + o(x^n)$$

$$\log(1+x) = x - \frac{x^2}{2} + \cdots + (-1)^{n-1}\frac{x^n}{n} + o(x^n)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^m \frac{x^{2m+1}}{(2m+1)!} + o(x^{2m+2})$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \cdots + (-1)^m \frac{x^{2m}}{(2m)!} + o(x^{2m+1})$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2m+1}}{(2m+1)!} + o(x^{2m+2})$$

$$\cosh x = 1 + \frac{x^2}{2} + \frac{x^4}{4!} + \cdots + \frac{x^{2m}}{(2m)!} + o(x^{2m+1})$$

$$\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \cdots + \left|\binom{-\frac{1}{2}}{m}\right| \frac{x^{2m+1}}{2m+1} + o(x^{2m+2})$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^m \frac{x^{2m+1}}{2m+1} + o(x^{2m+2})$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + \cdots + \binom{\alpha}{n}x^n + o(x^n)$$

$$\frac{1}{1+x} = 1 - x + x^2 - \cdots + (-1)^n x^n + o(x^n)$$

$$\sqrt{1+x} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + o(x^3)$$